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ORIGINAL ARTICLE

A hybrid feature selection framework collaborating image processing, evolutionary intelligence, and machine learning for covid-19 disease classification

Law Kumar Singh[1] · Munish Khanna[2] · Amit Yadav[3] · Rekha singh[4]

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Abstract

The timely identification of COVID-19 is essential to mitigate elevated mortality rates and prevent the future proliferation of the pandemic. Diagnostic test kits identify the illness; however, they often require considerable time and may present challenges regarding accuracy. Chest computed tomography (CT) testing demonstrates greater accuracy and can also serve as a diagnostic tool for COVID-19. This empirical investigation utilizes a three-step innovative and highly effective hybrid methodology that integrates image processing, soft computing, and machine learning techniques to detect COVID-19 from chest CT scans. Following the pre-processing of the chest CT images, we proceeded to extract 213 features categorized into various classes. During the second phase, the selection of the most informative features essential for prediction was conducted utilizing three soft computing algorithms: the grey wolf optimization algorithm (GWO), the salp swarm-based optimization algorithm (SSA), and a proposed hybrid approach combining both algorithms. Leveraging the features identified through soft computing algorithms, the five benchmark machine learning models were trained and utilized to classify these CT images into COVID-19 and non-COVID categories based on the chosen feature set. We undertook comprehensive testing involving 24 distinct tests (utilizing both 5-fold and 10-fold cross-validation approach). For each test, we evaluated performance across nine different standard metrics. The proposed model underwent evaluation using a publicly available CT dataset, achieving an impressive AUC of 0.9999 and a notable maximum accuracy rate of 96.90%. The findings from the experiments indicate that the suggested prediction approach surpasses other contemporary leading techniques. Expert radiologists facing heavy workloads may find the results of the proposed medical decision support framework beneficial as an additional perspective. The proposed solution is also particularly beneficial in areas experiencing a dearth of experienced medical practitioners. Additionally, we expect the suggested clinical decision system to be generalizable, and by choosing the most influential features for predictions of other human diseases, the same performance might be replicated on other datasets of human diseases.

Keywords Hybrid model · Salp swarm optimization algorithm · Grey wolf optimization algorithm · COVID-19 detection · Feature extraction · Feature selection

✉ Munish Khanna
munishkhanna.official@rocketmail.com

[1] School of Computer Science and Engineering, IILM University, Gurugram 122003, Haryana, India

[2] School of Computer Science and Engineering, Galgotias University, Gautam Budhh Nagar, Greater Noida 203201, Uttar Pradesh, India

[3] Department of Computer Science and Engineering, Anand School of Engineering and Technology, Sharda University, Agra 282007, Uttar Pradesh, India

[4] Department of Physics, Uttar Pradesh Rajarshi Tandon Open University, Prayagraj, Uttar Pradesh, India

1 Introduction

The rapid dissemination of COVID-19 has significantly impacted the physical and mental health of individuals globally. The COVID-19 virus was first identified in Wuhan, China, in December 2019. The virus subsequently disseminated from China to multiple countries, leading to rapid proliferation (Fig. 1). This situation has driven people to restrict their home activities and has thrust nearly every government and region worldwide into a state of emergency. Goel and colleagues, 2021. On July 27, 2020, the World Health

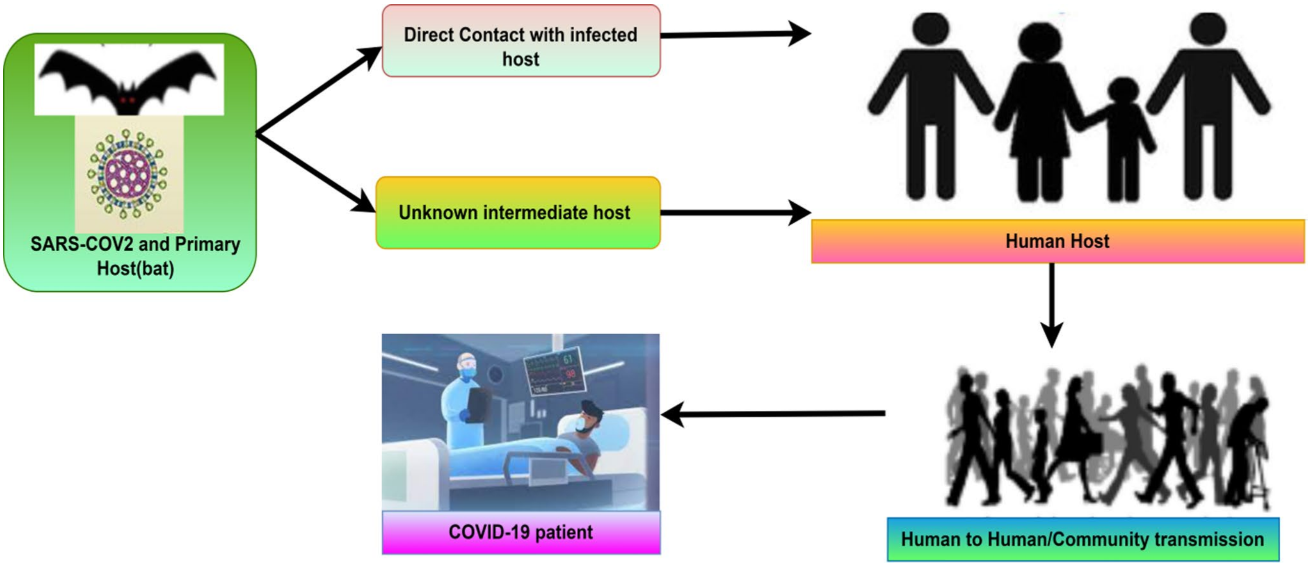

Fig. 1 Pictorial representation of SARS-CoV-2 transmission and spreading

Organization documented a cumulative total of 16,114,449 COVID-19 cases and 646,641 fatalities, subsequently declaring COVID-19 a public health emergency on January 30, 2020. Common indicators of COVID-19 include fever, cough, fatigue, difficulty breathing, and a reduced ability to taste and smell. Research suggests that artificial intelligence (AI) serves as an effective strategy in the fight against the COVID-19 virus. Mohamadou et al. 2020 provided a comprehensive review of COVID-19, encompassing various medical imaging studies, case reports, and treatment alternatives. Singh et al. 2020 emphasize the significant importance of chest X-ray (CXR) and CT scan images in the precise diagnosis of this condition. A number of radiologists and researchers have suggested the use of chest X-ray images for the diagnosis of COVID-19, given that the majority of radiological laboratories and hospitals are equipped with X-ray technology, facilitating the acquisition of patient chest images (Basavegowda and Dagnew 2020). Nevertheless, these CXR images are unable to differentiate between soft tissues (Tingting et al. 2019). To address this challenge, scholars favour CT-scan images that effectively identify soft tissues. A multitude of research efforts (such as Fang et al. 2020) have demonstrated the efficacy of CT imaging in accurately identifying individuals affected by COVID-19.

From a different perspective, certain patients infected with COVID-19 encounter medical complications, including acute respiratory conditions and secondary infections. Timely intervention following a diagnosis markedly decreases an individual's mortality risk in critical circumstances. At present, the gold standard for detecting COVID-19 illness involves a test known as reverse transcriptase-polymerase chain reaction (RT-PCR), which is conducted on swab samples obtained from the respiratory tract. The RT-PCR test is crucial; however, it involves a manual process that is time-consuming, complex, and requires significant labor. Moreover, the accessibility of testing kits remains constrained in numerous nations. Establishing a swift and dependable automated system for the early detection of COVID-19 is essential. CT imaging serves as a dependable method for identifying COVID-19, providing valuable support to the RT-PCR testing approach. The advancement of automated systems for CT image-based detection and diagnosis is being driven by AI, aiming to supplant the labor-intensive and error-prone manual testing for COVID-19 detection, while also easing the workload on specialized medical professionals.

1.1 Problem definition

Given that there is currently no perfectly targeted vaccine available for COVID-19 infection, and no medicine has shown to significantly help in clinical trials, timely finding of COVID-19 illness is significant for therapy and management of this disease. Without a doubt, the management of COVID-19 will put a lot of pressure on healthcare facilities (Wang et al. 2020a, b, c). Furthermore, front-line healthcare professionals are strangely infected by COVID-19 because of the lack of appropriate personal protective equipment. Currently, the quick identification and isolation of ill individuals serve as an effective method to prevent the healthcare system from overworking. Clinical systems' capabilities would be at risk without preventive measures. A disruption or complete breakdown of the healthcare system would lead to a significant fatality rate because all illnesses would go untreated. Detecting COVID-19 has become a time-consuming task, causing concern for everyone due to its limited accessibility (Khanna et al. 2023). The lack of COVID-19 confirmation kits, specifically in undeveloped countries, necessitates the adoption of other investigative approaches (Li et al. 2020a). Speedy and correct COVID-19 diagnosis is becoming increasingly crucial due to the potential for infected people to remain undetected and untreated. Infected individuals spread COVID-19 to healthy individuals because the virus is infectious. Despite the recent release of several COVID-19 diagnostic approaches based on data mining and AI (Li et al. 2020a), the COVID-19 outbreak curve remains unflatten. This empirical research objective is to develop a COVID-19 identification clinical system that inherits the advantages of effective image processing followed by feature selection (FS) through a meta-heuristic approach and classification using a ML approach.

1.2 Objective

Numerous nations lack, or experience a shortage of, the essential medical resources required to identify COVID-19. This indicates that an efficient and cost-effective alternative approach is required to detect and accurately identify the virus. Machine learning (ML) techniques have the potential to assist in identifying COVID-19 patients through the visual analysis of chest CT-scan images. When conducting multiple tests, it is essential to evaluate the resulting images with accuracy and efficiency. In rare instances, COVID-19 leads to bilateral pulmonary parenchymal ground-glass and consolidative opacities, noted for their rounded morphology and peripheral localization within the lungs. This research seeks to effectively pinpoint critical imaging features derived from chest CT scans. Over the past ten years, artificial intelligence (AI), particularly ML, has demonstrated considerable reliability within the medical sector, as evidenced by numerous previously published studies. Employing machine learning to forecast COVID-19 in patients could significantly decrease the turnaround time for medical test results, enabling healthcare professionals to administer the most appropriate treatment promptly.

The primary aim of this empirical investigation is to create a three-phase model utilizing image processing and artificial intelligence to ascertain whether a patient is infected with COVID-19, based on a defined set of optimal features(attributes). As a result, this research presents a hybrid model for detecting COVID-19 that focuses on optimizing CT-scan image features through SSA, GWO, and their successfully integrated combined optimization approach. The results of the chosen methodologies are evaluated to determine which approach exhibits the greatest level of performance. Essential features(characteristics) are selected from initially extracted features to evaluate their impact on classification. Another objective is to determine the most effective machine learning model(s) for prediction, applying these models to clinical reports of patients to achieve precise forecasts of COVID-19, and additionally to share the optimal set of features with the research community.

1.3 COVID-19 diagnose methodologies

COVID-19 can be diagnosed using one of three methods: (1) Real-Time reverse transcriptase-Polymerase Chain Reaction (RT-PCR) (Li et al. 2020b (2) chest CT imaging scan (Kovács et al. 2021), and (3) numerical (Brinati et al. 2020). RT-PCR assays are relatively rapid, sensitive, and consistent (Fig. 2). The test is taken from a person's throat or nose, and chemicals are used to remove any proteins, lipids, or other substances, leaving just the existing Ribonucleic Acid (RNA) (Huang et al. 2020). The separated RNA is a combination of a person's RNA and, if present, the RNA of the coronavirus. Despite its ubiquity, the RT-PCR test has the danger of producing false-negative and false-positive findings (Chen et al. 2020).

1.4 COVID-19 diagnosis techniques based on artificial intelligence

ML is a branch of AI that refers to software capable of autonomously determining how to perform a task or address a problem (Singh and Khanna 2022). The same principle underpins ML methods, which enhance their performance over time. Initially, ML models receive a set of training data, and on the basis of that, models construct the mathematical model using this training input data. After that, these mathematical models are used and implemented in the present context. Researchers have created a variety of identification algorithms for COVID-19 using ML techniques (Ahemad et al. 2022; Rahman et al. 2022). They have suggested a variety of ML-based approaches for the classification of COVID-19 through the analysis of chest CT images. For example, Shahin et al. 2022 categorize COVID-19 based on clinical and laboratory parameters. They employed support vector machine (SVM) and radial basis functions for classification. Wang et al. 2021 used a decision tree along with the Adaboost algorithm and combined it with a DL method to get the feature representation of chest CT images. They then use the features they got to classify COVID-19. Furthermore, Xu et al. 2020 uses an end-to-end network to transform CT images into label space for the identification of COVID-19. The majority of ML methods currently in use for COVID-19 diagnosis rely on either manually crafted features or visual representations learned through deep neural networks. It is also observed that researchers have

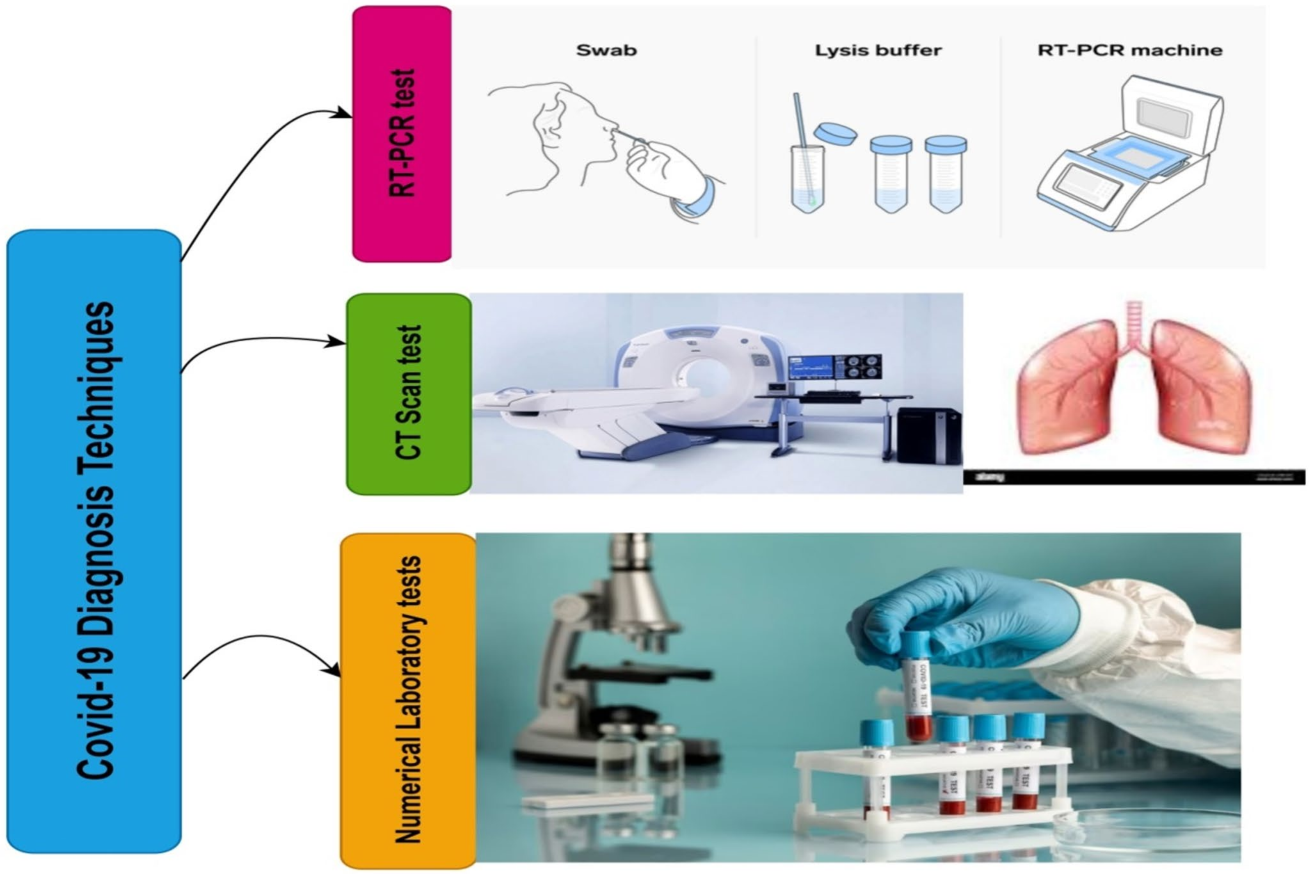

Fig. 2 COVID-19 testing techniques

also suggested using deep learning (DL)-based methods for COVID-19 diagnosis (Salama et al. 2024; Aggarwal et al. 2022a, b; Asif et al. 2024; Aravinda et al. 2025).

This study presents an innovative approach to FS, utilizing the exploration and exploitation of soft computing techniques to effectively select features from a moderate dataset of medical images for the categorization into COVID-19-infected individuals and healthy individuals. FS minimises the feature subset by eliminating unnecessary and redundant features, hence improving classification performance and enhancing efficiency. The proposed clinical decision classification system, whose core is the suggested FS approach, analyzes chest CT images to effectively distinguish between infected and healthy individuals. Initially, we used chest CT scans to extract 213 features using custom-developed software application so that initial feature vector can be created. Consequently, we developed a FS approach aimed at identifying the subset of most influential features and minimizing the dimensionality of the feature vector. In this phase, we employed soft computing algorithms named as SSA algorithm, GWO algorithm, and a proposed hybrid of these two algorithms so as to create the minimal subset of the most informative features from the initial original features set extracted from pre-processed chest CT scans benchmark dataset. Five different ML classifiers sequentially receive the selected set of features to make predictions. We calculated six efficiency indicators to assess the efficiency of the presented approach.

Although a number of hybrid soft computing-based models for COVID-19 detection have been put forth, many of them still have significant drawbacks that serve as the driving force behind this work. The majority of current hybrid approaches, particularly when working with high-dimensional COVID-19 datasets, frequently pay little attention to the quality, stability, and clinical relevance of the chosen features in favour of increasing classification accuracy through the combination of multiple algorithms. Hybridization is frequently used at the classifier level as opposed to the FS stage, which can result in redundant features, higher computational costs, and less interpretability. Additionally, a number of studies increase the search space and restrict generalization across datasets by relying on a single optimization stage without prior statistical filtering. The practical reliability of these models for real-world screening is further limited by their limited validation across multiple classifiers and their lack of emphasis on sensitivity and false-negative reduction. By combining complementary soft computing techniques to improve feature efficiency, robustness, and consistent predictive performance for COVID-19 detection, the suggested hybrid feature selection framework directly addresses these constraints. Specifically, the existing systems have issues such as increased architectural complexity, high computational cost, limited generalization capability across diverse datasets, and challenges in balancing feature extraction between fused models which has been addressed by the suggested approach.

Here are concise descriptions of the proposed work's main contributions:

1. This study introduces a robust hybrid three-phase model for predicting COVID-19 illness by integrating image processing techniques, soft-computing intelligence for feature selection, and ML for categorization.
2. Our clinical decision classification system employed three soft computing algorithms to identify most informative features while eliminating redundant and ineffective ones simultaneously. This reduced the computational time and training complexity required for developing a ML models. Therefore, the suggested model allows us to possess efficient features(characteristics).
3. To the best of our knowledge, we are the front runners to perform such type of thorough and innovative investigation of its kind, extracting a diverse array of features from various categories from CT images. Furthermore, to the best of our knowledge, we are pioneering the application of three soft computing algorithms to enhance features derived from CT images related to this infection, as well as employing five ML classifiers to identify eight critical metrics for evaluating performance.
4. We evaluated the effectiveness of the proposed method using a public dataset of medium-sized chest CT images. We assessed the model's effectiveness using various statistically significant performance measurement parameters, enabling a comparison with several recent approaches. The findings indicated that the suggested automated classification approach for COVID-19, which utilized handcrafted features, occasionally also outperformed deep learning (DL) models in few key efficiency measuring parameters like accuracy.
5. The proposed approach significantly reduces detection time while ensuring auspicious accuracy, which is a considerable advantage for clinical applications requiring timely results. Despite the limited number of image samples, thorough and meticulous testing indicates that employing multiple classification algorithms in conjunction can lead to highly accurate classification of COVID-19 data. We can advance this empirical work by developing a comprehensive web-based detection system that functions as a healthcare pipeline, facilitating the screening of potentially hazardous cases.

The rest of the study's subsequent sections follows this structure: Sect. 2 outlines the relevant background, while the second part 3 details the proposed methodology. Section 4 discusses the experimental results, presenting them in both graphical and tabular formats. Section 5 contrasts this study with previous high-quality state-of-the-art research. Section 6 encapsulates the conclusion, limitations and proposes several avenues for future exploration.

2 Related background

A multitude of studies in the scientific literature have documented the application of ML and Deep Learning (DL) techniques in X-ray and chest CT imaging for the detection of COVID-19. Recently, numerous researchers globally have introduced image analysis techniques utilizing ML and DL to predict this condition, thereby assisting doctors in achieving precise diagnoses.

The authors utilized a ResNet model to forecast the outcomes for COVID-19 patients (Zhang et al. 2020). They evaluated the performance of the proposed model using a dataset that comprised 1078 patient CXR images. The study by Wang et al. 2020a, b, c involved the analysis of CXR images from patients with COVID-19, pneumonia, and healthy individuals using CNN. Narin et al. 2021 utilized three prominent deep learning models—ResNet50, Inception-V3, and Inception-ResNetV2—on chest X-ray images. Their system identifies anomalies in an image through the use of class borders within the class decomposition technique. Maghdid et al. 2021 demonstrated the effectiveness of the model based on DL for the screening of COVID-19 disease. Rehman et al. 2022 improved recognition performance by using pre-trained weights and a transfer learning method. This made it possible to compare their results to those of other CNN models. Sahlol et al. 2020 initially introduced the hybrid classification approach. Researchers utilized a CNN model to extract features from the CXR images, followed by the application of the Marine Predators Algorithm to pinpoint the most significant traits for effective recognition.

Further discussing some recent state-of-the-art work recently published while focusing on predicting COVID-19 patients using chest CT imaging, Tang et al. 2020 emphasized the importance of chest CT images in assessing the severity of COVID-19 infection. The researchers introduced a technique utilizing ML to assess the severity of COVID-19 through chest CT imaging analysis. Gopatoti and Vijayalakshmi 2022 presents a three-stage classification model for COVID-19, which utilizes CXR images, DL convolutional neural networks, and a sophisticated FS technique known as the enhanced grey-wolf optimizer with genetic algorithm. He et al. 2020 introduced the Self-Trans method, focusing on delivering robust and unbiased features. This technique merges strategies for self-directed learning with the utilization of knowledge across various settings. Loey et al. 2025 initially detailed their progress in enhancing data and forecasting COVID-19 using a deep transfer learning model and conditional generative adversarial networks (CGAN). Sahin 2022 introduced an innovative Convolutional Neural Network (CNN) model in their study for the automated detection of COVID-19 through chest X-ray images. The suggested CNN model for binary classification aims to serve as a reliable diagnostic tool for distinguishing between normal and COVID cases. They assess various architectures alongside the proposed model for this COVID-19 dataset, including the pre-trained MobileNetv2 and ResNet50 models. Zhang et al. 2023 presented a ML-powered COVID-19 prediction model that combines the k-nearest neighbor (KNN) classifier with an enhanced whale optimization strategy. The proposed approach was fundamentally based on the random contraction strategy and the Rosenbrock method. For predicting COVID disease, Al-Qaness et al. 2020 used an innovative ANFIS (adaptive neuro-fuzzy inference) approach. Employing Genetic algorithm and enhanced K-Nearest Neighbor classifier, the proposed method Shaban et al. 2020 introduced an innovative approach for confirming COVID-19 cases through their newly developed COVID-19 Patients Detection Strategy method. Practitioners created a combined approach for selecting features to identify the key characteristics derived from chest CT scans of both non-COVID-19 and COVID-19 patients. Barstugan et al. 2020 proposed an approach that uses ML algorithms to classify COVID-19 patients. Five extraction methods utilized the feature mining technique on CT images to accurately identify the affected regions. Subsequently, the SVM classification method sorted the patients based on their acquired characteristics. The results of the experiments showed that the grey level size zone matrix worked as expected when it was combined with SVM. Aggarwal et al. 2022a, b conducted a comparison of fine-tuned DL models aimed at enhancing the detection and classification of COVID-19 patients in relation to other types of pneumonia. The addition of a new set of network head layers has improved the performance of the eight CNN models, as proposed in the study.

Wang et al. 2021 predicted that the DL algorithm would be capable of recognizing the distinctive graphical characteristics of COVID-19 to aid in clinical diagnosis. The findings presented by the practitioners indicate that graphic element extraction enables DL algorithms to identify COVID-19 patients. Farid et al. 2020 noted that research on COVID-19 utilized a probabilistic approach alongside a classification system. This approach enables the classification of COVID-19 patients according to the prominent characteristics observed in their CT scans. Prior to implementing the recommended hybrid classification approach, they gathered and refined features from CT scans. Hemdan et al. 2020 presented a proposed architecture that uses pre-trained deep learning classifiers (COVIDX-Net) to automatically diagnose COVID-19 using X-ray images. Seven distinct configurations of CNN models were required to apply the suggested approach. Within the context of COVID-19 screening, the researchers, Kathamuthu et al. 2023, examined chest CT images utilizing various CNN models. The VGG16 model achieved an impressive accuracy rate of 98%, making it the most precise among the various models evaluated (like VGG16, VGG19, Densenet121, InceptionV3, Xception, and Resnet50).

Many recently published studies focussed on the importance and the impact of efficient feature selection process. This recently published study classified breast cancer histopathology images using an adaptive penalized logistic regression model and the weighted L1-norm (Alkahya et al. 2023). Classifying these images requires selecting the best image features from thousands of candidates. The study proposed a compatible logistic regression-weighted L1-norm method to identify features. Our method was tested against state-of-the-art methods using publicly available breast cancer histopathological image datasets. The method outperforms three others in classification accuracy and FS. In terms of selected features, the proposed method outperforms others. The results show that the proposed FS method for medical image classification is promising. Support vector classification (SVC) is a popular classification method. Feature selection, kernel parameter, and penalty parameter settings significantly impact SVC classification performance. The grey wolf optimization (GWO) algorithm was employed to improve FS and determine SVC parameter values simultaneously in this published empirical study(Qasim and Algamal 2021). FOGWO-SVC can select the best features with best parameters, according to experimental results from several disease benchmark datasets. Comparative results show that FOGWO-SVC outperforms competitor algorithms in classification accuracy and feature reduction. Previous researchers selected features using binary nature-inspired algorithms. Each algorithm needs an initial population, and the initialization is crucial to the result. Positions are initialized randomly by uniform distribution during population initialization, resulting in high classification variability. Binary nature-inspired algorithms use parametric and non-parametric methods like the t-test and Wilcoxon rank sum test as initial populations to avoid randomness and account for the relationship between each feature and the class variable. With this modification, these binary algorithms can improve global exploration and local exploitation or converge slower than before(Al-Thanoon et al. 2022). Researchers examined binary bat, grey wolf, and whale algorithms. The proposed methods were tested on ten publicly available high-dimensional and low-dimensional datasets. Experimental results and statistical analysis show that the methods outperformed standard algorithms in classification accuracy, FS stability, running time, and FS number. Swarm algorithms are used in FS and require transfer functions to change search space from continuous to discrete. However, transfer functions underpin all binary swarm algorithms. Binary swarm algorithms cannot balance exploration and exploitation with transfer functions in the current formula. The binary whale optimization algorithm with various updating methods for time-varying transfer functions was used to select features in this published work(Kahya et al. 2020). Three chemical and biological binary datasets were used to test the proposed method. The results showed that BWOA-TV2 has consistent FS, which increases classification accuracy and convergence. The proposed method outperformed particle swarm optimization (PSO) and firefly optimization (FO), which are commonly used in this field. The published study, (Ewees et al. 2023) improved the slime mould algorithm to create a new FS method. The proposed local search method uses the marine predators algorithm (MPA)'s exploration ability. They present SMAMPA, a modified Slime mould algorithm(SMA) that uses Marine Predators Algorithm (MPA) operators as a local search strategy to increase convergence rate and avoid local optima. Using multiple datasets, SMAMPA was compared to state-of-the-art FS methods for efficiency. The results show that the developed SMAMPA method can reduce dataset size by increasing prediction rate.

A significant challenge in ML is the high dimensionality of datasets, necessitating substantial memory for the analysis of numerous features, which results in overfitting. Consequently, the FS process minimizes redundant data and processing time, thereby enhancing the algorithm's performance and this limitation is also addressed in this work. Moreover, there is always a demand for expert medical practitioners required for the diagnosis of this disease. Especially in developing countries and rural areas, these experts are not easily available; moreover, there is always a chance of intra-observer variability among experts while taking decision on potential patient is really diseased or not. Also, the in-depth analysis of the results calculated through the recent state-of-the-art clearly indicates that there is still scope for improvement in the results of different performance-measuring parameters (like accuracy, sensitivity, AUC, F1-score and specificity). Thus, this study may fill up these all three gaps and side-by-side playing the role of second opinion and reducing the load of expert medical practitioners at the same time.

Next, the literature review has been expanded to explicitly highlight the existing wok in current FS research for COVID-19 classification. Most prior works rely either on purely ML based classification techniques, which may overlook nonlinear dependencies, or on standalone soft computing algorithms, which often suffer from high computational cost and instability in medical datasets. Very limited studies systematically integrate multiple soft-computing algorithms (covering exploration and exploitation both in-depth) while evaluating their impact across multiple classifiers and diseases separately. This gap directly motivates the proposed hybrid framework, which aims to achieve a balanced trade-off between interpretability, robustness, and classification performance. This structured methodological design ensures that the conclusions drawn are logically sound, clinically relevant, and supported by empirical evidence, thereby strengthening the overall contribution of the research.

We've reviewed over 110 state-of-the-art studies from previous years to understand the trajectory of the research community and to explores various innovative research efforts focused on predicting COVID-19 through the analysis of X-ray images and thoracic CT scans. Researchers are meticulously examining chest X-rays and CT scans to develop automated algorithms for predicting COVID-19 cases. Methods based on ML, Deep Learning (DL), and transfer learning have significantly propelled the progress of computer-assisted COVID-19 screening. However, there has been limited research on the extraction of features from CT scans of suspected patients and the application of soft computing techniques to identify the most significant ones. A variety of researchers have developed techniques rooted in DL that involve comparing images with previous models to identify this condition. Several scholars have demonstrated creativity in their work by developing a novel DL model, proposing an innovative ensemble strategy, offering hyper-parameter tuning insights, or providing recommendations for enhancing existing models. Researchers have utilized the properties of DL models, also known as deep features, to instruct ML models for classification purposes. This presented study is one of the front-runners and appears to be the unique study of its kind to explore a novel strategy for predicting this disease. This process extracts image processing features from various classes and sends them to three soft-computing algorithms designed to classify this disease. We then provide the most informative features selected through these algorithms to various ML classifiers to evaluate their effectiveness. Furthermore, our analysis revealed that the scale of our subject dataset and the quantity of obtained features significantly surpass those of previously published high-quality studies.

Finally, the comparative discussion has been enhanced by including a more comprehensive analysis against recent and well-established methods reported in the literature. The proposed hybrid FS method is compared with two baseline soft-computing algorithms but also with recent state-of-the-art approaches (and FS techniques). The comparison is performed using consistent dataset, cross validations, and uniform evaluation metrics such as accuracy, sensitivity, and specificity. This ensures fairness and allows a clearer assessment of how the proposed method improves diagnostic reliability, particularly in terms of reducing false negatives, which is critical in medical decision making.

Eliminating elements that hinder classification can enhance the model through strategies based on feature selection (FS). As a result, the model requires reduced training time, leading to improved accuracy in the classifications made by the learning model. This real-world study shows a three-step classification model that uses standard feature extraction algorithms to get features and a soft-computing algorithm-based FS method to find the most significant features for better COVID-19 prediction. We forecasted instances that could be bi-stage hybrids of COVID-19 by utilizing the SARS-CoV-2 CT scan images along with the COVID-CT database. Figures 3, 4 and 5 illustrate how the anticipated model operates. At the beginning of our process, our tailored Python and MATLAB programs serve as feature extractors. We use a benchmark, easily and freely downloadable SARS-CoV-2 CT scan dataset, comprising 1252 positive CT scans for COVID-19 and 1230 non-infected CT scans from SARS-CoV-2 patients, for a total of 2482 CT scans (PlamenEduardo 2020; Soares et al. 2020). The data presented here comes from real patients receiving care in hospitals in Sao Paulo, Brazil. This database aims to enhance research and development in systems capable of determining an individual's infection status with SARS-CoV-2 through image analysis. A soft computing-based FS approach selects the most pertinent characteristics from a vast array of features. To use the most vital subset of features, we employ the resulting feature set selected using two nature-inspired algorithms and their hybrid approach. For both COVID and non-COVID cases, various benchmark ML classifiers utilize this influential set of features to analyze chest CT images.

3 Methods and material

3.1 Feature extraction and selection

The chest CT images were obtained from benchmark online sources, so acquisition parameters (e.g., resolution, scanner type, slice thickness) may not fully standardized. The expected heterogeneity reflects real clinical settings but may introduce variability in image features and affect model performance. To reduce its impact, preprocessing steps such as resampling, intensity normalization, and data augmentation (along with transforming the original image into greyscale, subsequently implementing the CLAHE procedure, which is a form of adaptive histogram equalization (AHE),)were applied to improve robustness and generalization. Augmentation methods such as rotation, flipping, scaling, and intensity variations were used to increase the effective size of the minority class. In addition, balanced sampling strategies were employed during training to reduce bias toward the

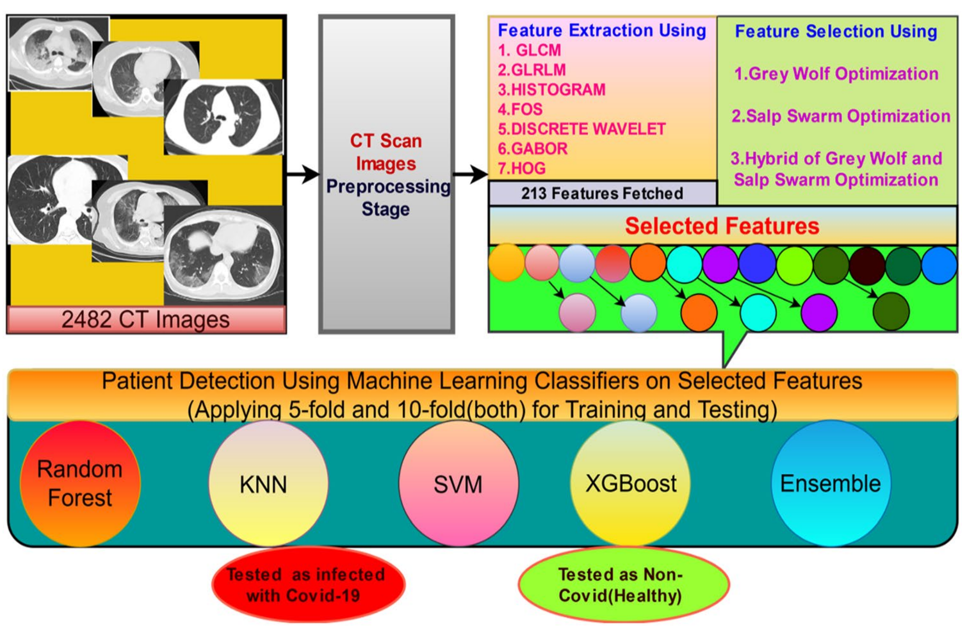

Fig. 3 Clinical decision classification system proposed in this study for COVID-19 identification

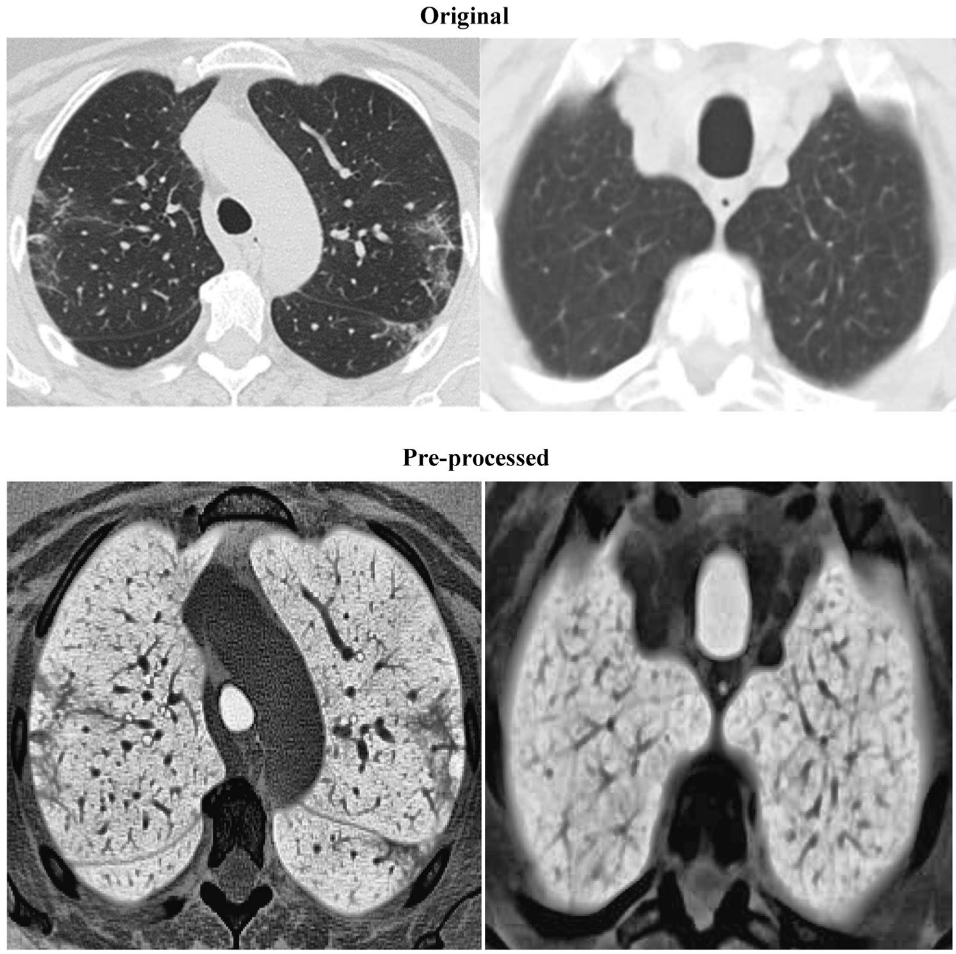

Fig. 4 Original and the same image after preprocessing selected from dataset

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Initialize the grey wolf population x_j (j =1,2,……………………………………… l)
Initialize χ , p and cv
Calculate the fitness of each search agent
$x_{\chi 1}$ =The best search agent
$x_{\chi 2}$ =The second best search agent
$x_{\chi 3}$ =The third best search agent
while(s < Max number of iterations)
    for each search agent
    Update the position of the current search agent by equations (7)
    end for
    Update χ , p and cv
    Calculate the fitness of all search agents
    Update $x_{\chi 1}, x_{\chi 2}$ and $x_{\chi 3}$
    $s = s + 1$
    end while
    return $x_{\chi 1}$

Input Parameters:
Datasets:**213*2482**

Define objective(cost) functions: $f(x)$

Initialize the salp population $P_l(l = 1, 2..................n)$ considering ub and lb

while (end condition is not satisfied)
    Calculate the fitness of each search agent(salp)
    B=the best search agent
Compute the value of d_1 by Eq. (9)
        for each salp(P_l)
                if($l == 1$)
                Update the position of the leading salp by Eq.(8)
                Else
                Update the position of the follower salp by Eq.(10)
                End
        end

        Compute the fitness of new population using a cost function, f(x)
end
return F

majority class and ensure more reliable performance across both classes.

We have identified and then segregate the extracted features into 7 distinct classes of features. Initially, we have obtained a grey level co-occurrence matrix (GLCM) feature, which fundamentally involves statistical techniques. This examines the relationships between pixels in a spatial context through the use of the grey-level co-occurrence matrix or the grey-level spatial dependence matrix. We calculate the total by counting the number of times the pixel value appears in the input image at a defined distance from a given pixel. This matrix contains information about textural features. This approach involves assessing the texture of an image by evaluating the occurrence of paired pixels that possess specific values within the image. We extracted different features, like contrast (contr), correlation (corrp), autocorrelation (autoc), cluster prominence (cprom), cluster shade (cshad), dissimilarity (dissi), and more, once we have the GLCM matrix. Maximum probability, Sum of squares variance, Energy, entropy, homogeneity, the information measure of correlation2, the sum average, the sum variance, the difference entropy, the difference variance, the inverse difference normalized, and the inverse difference moment normalized are all included. After obtaining the GLCM matrix, we proceed to compute various features such as short run emphasis (SRE), long run emphasis (LRE), grey level run emphasis (GLN), run percentage (RP), and run length non-uniformity (RLN). In summary, we calculated a total of 26 features from the gray-level co-occurrence matrix, 11 features from the gray-level run-length matrix, 32 histogram features, 16 features from the shape feature, 16 discrete wavelet features, 16 Gabor features, and 96 features from the histograms of oriented gradients.

A strategy for extracting features within the feature space can lead to the generation of redundant or superfluous features. It is essential to remove duplicate and unnecessary features(attributes), as this enhances classification accuracy and reduces computational expenses. The feature selection (FS) process generally accomplishes this by selecting the subgroup of selected features that exhibit optimal performance in a particular classification system. Through the application of statistical or probabilistic methods or soft-computing based methods, these filtering techniques evaluate and rank the features. The classification model either chooses or discards the feature based on the assigned scores. ML algorithms commonly employ techniques which use learning algorithms to highlight the importance of features and identify the most influential feature subsets for improving classification performance. Conversely, an embedded model integrates FS within a specific ML approach. The proposed study employed a soft-computing strategy to identify the most critical feature set for developing an almost perfect classification system for COVID-19 prediction on the basis of patients CT scan images (refer Figs. 3 and 5 for reference). The following sections will go into greater detail about the chosen and implemented approach.

Pseudocode GSSO-FS – Grey Wolf and Salp Swarm Optimization for Feature Selection

Input:
Dataset D with n samples and m features, Population size N , Maximum number of iterations T, classifier and balance weights ω_1 and ω_2 for classification error and feature size
Output:
Optimal feature subset f* with highest fitness
Step 1: Initialization
1.1. Initialize a population of N binary vectors Xi ∈ $\{0,1\}^m$ (randomly generated).
1.2. Divide the population into two groups:
- Salps (for exploration)
- Wolves (for exploitation)
1.3. Evaluate the fitness $J(X_i)$ of each solution using:
1.4. Identify the top three solutions as α (best), β (second-best), and δ (third-best).
Step 2: Main Optimization Loop (for t = 1 to T)
2.1. Compute exploration coefficient:
c1 = 2 - (2 × t / T)
Step 3: Salp Swarm (Exploration Phase)
3.1. For each salp si in the salp population:
a. If i == 1 (leader):
Update position:
si_j = Xα_j + c1 × ((ub_j - lb_j) × rand1 + lb_j)
b. Else (followers):
Update position:
si_j = (si_j + s(i-1)_j) / 2
Step 4: Grey Wolf (Exploitation Phase)
4.1. For each wolf wi in the wolf population:
a. Update each dimension j using:
- Compute distance vectors:
Dα = |C1 × Xα - Xi|
Dβ = |C2 × Xβ - Xi|
Dδ = |C3 × Xδ - Xi|
- Compute position influences:
X1 = Xα - A1 × Dα
X2 = Xβ - A2 × Dβ
X3 = Xδ - A3 × Dδ
- Update wolf's position:
Xi = (X1 + X2 + X3) / 3
where:
A = 2a × rand - a, C = 2 × rand, a = 2 × (1 - t/T)
Step 5: Elite Selection
5.1. Combine top p% of wolves and top q% of salps to form an elite pool.
5.2. Re-evaluate all elite solutions.
5.3. Update α, β, δ using best three solutions from elite pool.
Step 6: Termination
6.1. After reaching max iterations T, return the best solution Xα as the optimal feature subset f*.
End of Algorithm

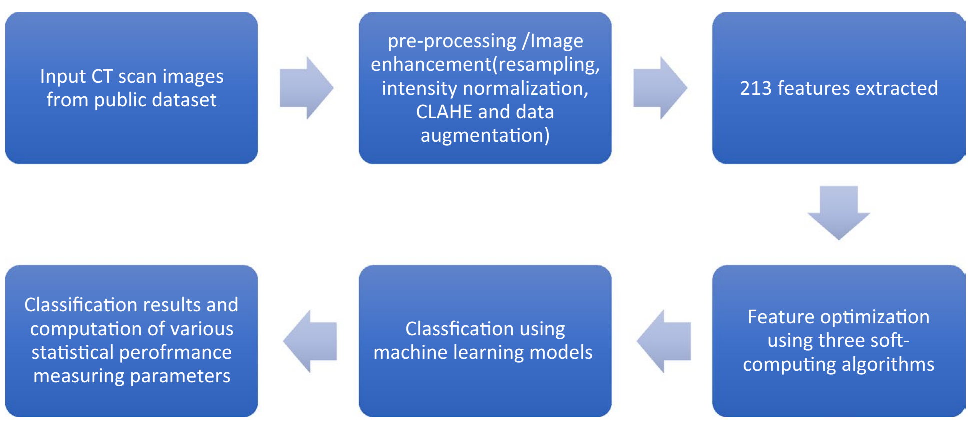

Fig. 5 Schematic Representation of COVID-19 Prediction System

Before the extraction of features from publically and easily approachable selected dataset, the initial step we implement consist of transforming the original image into greyscale, subsequently implementing the CLAHE procedure, which is a form of adaptive histogram equalization (AHE), to improve the contrast of the image (Fig. 4). Transforming a colour image into greyscale is essential for textural analysis.

3.2 Feature selection algorithms

Feature selection (FS) involves the selection of a reduced number of optimized features from a comprehensive set, aiming to preserve accuracy while ensuring system efficiency. The complexity of machine learning roles has led to the creation of a variety of methods to minimize redundant and superfluous data attributes. These algorithms are extremely effective that seek answers to complicated optimization problems, such as classical traveling salesman issues, scheduling problems, and profit maximization problems, among others. Evolutionary approaches like nature-inspired computing algorithms are a gathering of unique problem-solving techniques and strategies borrowed from natural processes. Popular examples of optimization algorithms inspired by nature include the genetic algorithm, artificial bee colony, particle swarm algorithm, and ant lion algorithm, among others. For the optimization under consideration, we've used the GWO algorithm and SSA algorithm.

3.2.1 Grey wolf optimization

Grey Wolf Optimization (GWO), proposed by Mirjalili et al. 2014; is a nature-inspired algorithm motivated by the hunting behavior and social hierarchy of grey wolves. The simulation features four distinct types of wolves within the leadership hierarchy. The wolves are named Alpha, Beta, Gamma, and Delta. Wolves locate their target, encircle it, and ultimately take it down during the hunting process. The dominant wolf has taken the group's lead, directed activities and assigned roles to other members. The alpha wolf holds greater significance than the beta wolf. The dominant wolf directs the subordinates, and the subordinates follow the dominant wolf's instructions. As omega wolves occupy the lowest tier in the food hierarchy, they must communicate with the higher-ranking wolves. Delta wolves don't fit into alpha, beta, or omega. They may serve as observers who monitor the pack's boundaries and alert others to potential dangers, or they may act as protectors who ensure the pack remains secure and unharmed. The GWO algorithm determines the optimal solution by creating a mathematical representation of the hunting behaviour of grey wolves. They pursue their target in a circular pattern. The mathematical equations that explain the behaviour of the GWO are discussed below.

The equations are:

$$\overrightarrow{D} = |\overrightarrow{cb}\overrightarrow{x_p}(s) - \overrightarrow{x}(s)| \tag{1}$$

$$\overrightarrow{x}(s+1) = \overrightarrow{x}_p(s) - \overrightarrow{P}\overrightarrow{D} \tag{2}$$

Where is s denote the current iteration, $\overrightarrow{P}$ and $\overrightarrow{cb}$ are the coefficient vector. $\overrightarrow{x}_p$ denote the position vector of the prey and $\overrightarrow{x}$ represent the grey wolf position vector.

The grey wolf is capable of adjusting its position within the surrounding area of the prey at any arbitrary location by employing Eqs. (1) and (2).

$$\overrightarrow{p} = 2\overrightarrow{b}\overrightarrow{a1} - \overrightarrow{b} \tag{3}$$

$$\overrightarrow{cb} = 2\overrightarrow{a2} \tag{4}$$

The vector $\overrightarrow{p}$ and $\overrightarrow{cb}$ are calculated using Eqs. (3) and (4).

$\overrightarrow{b}$ decreased linearly from 2 to 0. $a1$and$a2$are random vectors in [0,1].

$$\overrightarrow{D}_{\chi 1} = |\overrightarrow{cb}_1\overrightarrow{x}_{\chi 1} - \overrightarrow{x}|, \overrightarrow{D}_{\chi 2} = |\overrightarrow{cb}_2\overrightarrow{x}_{\chi 2} - \overrightarrow{x}|, \overrightarrow{D}_{\chi 3} = |\overrightarrow{cb}_3\overrightarrow{x}_{\chi 3} - \overrightarrow{x}| \tag{5}$$

$$\overrightarrow{x}_1 = \overrightarrow{x}_{\chi 1} - \overrightarrow{P}_1(\overrightarrow{D}_{\chi 1}), \overrightarrow{x}_2 = \overrightarrow{x}_{\chi 2} - \overrightarrow{P}_2(\overrightarrow{D}_{\chi 2}), \overrightarrow{x}_3 = \overrightarrow{x}_{\chi 3} - \overrightarrow{P}_3(\overrightarrow{D}_{\chi 3}) \tag{6}$$

$$\overrightarrow{x}(s+1) = \frac{\overrightarrow{x}_1 + \overrightarrow{x}_2 + \overrightarrow{x}_3}{3} \tag{7}$$

The settings for the parameters in the experimentation process of the proposed methodology are outlined as follows:

•Variations in the size of population	(5,10,15,20)
•Variations in the count of iterations	(100,200,300,400,500)
•Count of initial features extracted	213
•Count of considered instances	2482
•Lb (Lower Bound)	0
•Ub (Upper Bound)	1

3.2.2 Salp swarm algorithm (SSA)

The collective movement patterns of salps inspired the development of the SSA in 2017. To optimize the current solution, researchers introduce a new control parameter known as "inertia weight.". Mirjalili et al. 2017 introduced the Salp swarm algorithm (SSA), which effectively addresses a wide range of optimization challenges. The behavior aligns perfectly with the natural conduct of salps in their habitat. Salp is one of the salpide family's subfamilies, which includes planktonic tunicates with a barrel-like shape. Furthermore, their tissues resemble those of jellyfish, and a large amount of water contributes significantly to both their mobility and overall weight. SSA divides its employees into two distinct groups: leaders and followers. The salp at the forefront assumes the role of leader, whereas the others follow as subordinates. In a multidimensional search space, the goal of the salp is to locate the source of nourishment.

The leader's position is revised according to the subsequent equation.

$$P_k^1 = \begin{cases} F_k + d_1((ub_k - lb_k)d_2 + lb_k) & d_3 \geq 0.5 \\ F_k - d_1((ub_k - lb_k)d_2 + lb_k) & d_3 < 0.5 \end{cases} \tag{8}$$

where P_k^1 and F_k are the positions of leaders and food source k^{th} dimension, respectively.d_i is a variable that is systematically reduced throughout the iterations, as defined in Eq. 9, where i represents the current iteration and I denote the maximum number of iterations. The additional d_2 and d_3 variables in Eq. (8) represent two values selected at random from the range [0,1].

$$d_1 = 2e^{-\left(\frac{4i}{I}\right)^2} \tag{9}$$

$$P_k^l = \frac{1}{2}\left(P_k^l + P_k^{l-1}\right) \tag{10}$$

3.2.3 Hybrid algorithm model

A hybrid algorithm integrates multiple distinct methodologies to address a specific issue. People frequently execute this to combine the advantageous characteristics of each method, resulting in an algorithm that surpasses the mere combination of its components. The term "hybrid algorithm" specifically pertains to the integration of algorithms that tackle the same problem while differing in various aspects, particularly in terms of performance, rather than merely merging multiple algorithms to address distinct issues (Shrivastava and Khanna 2025). We can view hybrid algorithms as integrations of more basic algorithms. This empirical study presents a clear approach for reducing features set by employing two distinct and comprehensive algorithms that work together to improve the features selection from the original selected data set and side-by-side minimizing the use of computational resources as well. Additionally, numerous methods designed to reduce feature values often cater to particular datasets or goals. Nonetheless, integrating different optimization algorithms can improve accuracy and reduce the number of selected features for a specific dataset. The primary goal is to merge the key characteristics(behaviour) of each component(algorithm), resulting in a holistic approach that surpasses the abilities of the separate components. Combining various algorithms to address a particular problem is one approach to creating a hybrid algorithm. We can view a variety of algorithms as assemblies of smaller parts. Instead, it involves gathering algorithms that address the same issue but vary across different aspects, such as their effectiveness. GWO, SSA, and their hybrid were selected for FS due to their complementary optimization strengths and proven effectiveness in high-dimensional problems. GWO provides strong exploitation and fast convergence, making it suitable for refining optimal feature subsets, while SSA offers powerful global exploration, reducing the risk of premature convergence. The hybrid of GWO–SSA algorithms balances exploration and exploitation, a key theoretical requirement in metaheuristic optimization. Empirical studies have shown that these methods, especially in hybrid form, outperform single optimizers in FS by improving classification accuracy, reducing feature redundancy, and enhancing convergence stability, particularly in medical datasets.

Suggested hybrid approach for enhancing feature selection.

The proposed method integrates the GWO and SSA algorithms into a unified approach. GWO excels in exploitation (refinement near the best solutions) due to its social hierarchy model (α, β, δ, ω wolves) and encircling mechanism. SSO is better in exploration, using its leader–follower chain model to spread over the search space. The salp swarm method enhances the basic Grey Wolf optimization technique, leading to better convergence and efficiency.

Roles of each algorithm component are as follows.

1. **Exploration (SSA Component)**:

   The SSA component drives exploration through the leader's movement which incorporates large jumps (via c_1), the chain behaviour that allows wide sampling of search space and random coefficients c_2, c_3 that promote diversity.

2. **Exploitation (GWO Component)**:

   The GWO component provides strong exploitation through the hierarchical structure that focuses search around best solutions, the encircling behaviour that allows fine-tuning of solutions and linear decrease of 'a' parameter that gradually shifts to exploitation.

Exploration (SSO).

The leader salp is guided toward the best-known solution Xα:

$$S_i^j = X_\alpha^j + c_1.\ ((ub_j - lb_j).\ r_1 + lb_j)$$

Followers are updated via:

$$S_i^j = \frac{S_i^j + S_{i-1}^j}{2}.$$

Exploitation (GWO).

Wolf position update rule:

$X_i = \frac{X1+X2+X3}{3}$ where $X_k = X_k^{best} - A_k\,|C_k.\ X_k^{best} - X_i|$.

Fitness function.

Each solution is represented as a binary vector of length D (number of features) where 1 indicates the feature is selected and 0 indicates the feature is not selected. In the suggested approach for selecting features, the cost associated with each produced population is characterized as:

$$f(x_i) = -(0.99 * (1 - E(x_i)) + 0.01 * sum(S)/(N)) \qquad (11)$$

where$f(x_i)$represents the cost of $x_i.E(x_i)$ represents the performance value or classification accuracy of the ith feature set. Let S represent the count of chosen features, while N denotes the total number of features present in the original dataset.

Moreover, initial crossover probability is set to 0.7, a (GWO) decreases linearly from 2 to 0 and c_1 (SSA) is adaptive.

Let N be the population size, m be the number of features, T is the number of iterations and C the cost of fitness evaluation (classifier cost), then the overall time complexity can be computed as

O(T·N·(m+C)).

where O(m) is featuring vector update and O(C)is the fitness function (e.g., using 5-fold CV). If we consider.

cross validation approach then the time complexity can be considered as

O(T·N·(m+n·logn)).

There seem multiple advantages of the Hybrid Approach like combining GWO's local search intensity with SSA's global search capability (balanced search), the crossover probability decreases linearly, shifting from exploration to exploitation (adaptive behaviour), binary conversion and fitness function tailored for FS(specifically) and can handles high-dimensional data common in medical applications. The block diagram of the proposed clinical classification support system for fast and effective COVID-19 detection is depicted through Figure 5.

3.3 Extracted features

There are 213 features in total, one target label, or 214 columns. There are 2482 instances in total, of which 1252 are positive for SARS-CoV-2 infection (COVID-19) and 1230 are CT scans for patients who are not SARS-CoV-2-positive (Table 1).

3.4 Results and discussion

As previously mentioned, this empirical study involves two nature-inspired computing algorithms and introduces a novel approach which integrates them for feature selection and classification (utilizing machine learning) in the detection of COVID-19 infection. The mathematical fitness evaluator utilizes GWO, SSA, and their combined methodologies to ascertain the selection of features. In order to assess the chosen subset of features, we utilize Random Forest, KNN, SVM, XGBoost, along with a combination of these classifiers (ensemble model). Through these constructive initiatives, we anticipate to identify an effective approach for detecting COVID-19. In this pursuit, we seek to provide the research community with an innovative set of highly effective features and present a framework that can aid individuals in precisely identifying COVID within a substantial patient population. The approach employed leverages Python on a Ryzen 5 system featuring a 3.5 GHz CPU and 8 GB of RAM. For extraction of certain features MATLAB has also been used. As previously stated, we have implemented three bio-inspired evolutionary algorithms and performed eight tests for each algorithm, adjusting the population size to 5, 10, 15, and 20, respectively, with a range of 100 iterations from the initial 100 to 500 iterations. This research utilized both 5-fold and 10-fold cross-validation methods to evaluate the efficacy of the proposed approach using nine standard metrics for measuring efficiency. The analysis encompasses statistical parameters, ROC curves, and execution time, as detailed in Tables 2, 3 and 4. Alongside these assessments, we have incorporated an AUC, confusion matrix visualizations, a graph illustrating the correlation between fitness and iterations, as well as a table outlining the number of features and their respective counts for experiment 2 of the hybrid method. Figures 6 and 7, and 8 illustrate the selected convergence curves (Iteration versus Fitness). Due to limitations in space, we are unable to include convergence curves and tables for each experiment executed, but we can make them available upon request for interested readers. Additionally, we provide the optimal values derived from experiments 1–24 for the same

Table 1 Class of features and the number of features extracted from CT scans of dataset

Class of features	Number of features under particular class
Discrete Wavelet features	16
FOS	16
GLRM	11
HOG	96
GLCM	26
Histogram	32
Gabor Features	16

Table 2 Best results computed through features returned by GWO algorithms

Experiment -1 with population size 5 and 5-fold cross-validation									
Best value (Classifier Name)	**Specificity (Spcfty)**	**Sensitivity (Snstvvty)**	**Kappa Score(Kappas)**	**Precision (Prcson)**	**F1-Score**	**MCC**	**Accuracy (Accrcy)**	**AUC Score (AUCS)**	**Time in execution (Exctn-time)**
	0.9552 (XGBoost(XGB))	0.9559 (XGB)	0.9105 (XGB)	0.9544 (XGB)	0.9552 (XGB)	0.9105 (XGB)	0.9398 (XGB)	0.9910 (XGB)	1.84114 (KNN)
Experiment -2 with population size 5 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9560 (XGB)	0.9568 (XGB)	0.9129 (XGB)	0.9553 (XGB)	0.9564 (XGB)	0.9129 (XGB)	0.9447 (SVM)	0.9929 (XGB)	4.92636 (KNN)
Experiment -3 with population size 10 and 5-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9608 (XGB)	0.9568 (XGB)	0.9177 (XGB)	0.9600 (XGB)	0.9588 (XGB)	0.9177 (XGB)	0.9439 (RF)	0.9946 (XGB)	5.04537 (KNN)
Experiment 4- with population size 10 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9560 (XGB)	0.9568 (XGB)	0.9129 (XGB)	0.9553 (XGB)	0.9564 (XGB)	0.9129 (XGB)	0.9447 (SVM)	0.9929 (XGB)	4.92636 (KNN)
Experiment 5- with population size 15 and 5-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9568 (XGB)	0.9544 (XGB)	0.9113 (XGB)	0.9559 (XGB)	0.9556 (XGB)	0.9113 (XGB)	0.9443 (SVM)	0.9919 (XGB)	3.16023 (SVM)
Experiment 6- with population size 15 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9600 (XGB)	0.9552 (XGB)	0.9153 (XGB)	0.9590 (XGB)	0.9576 (XGB)	0.9153 (XGB)	0.9496 (XGB)	0.9935 (XGB)	5.93444 (KNN)
Experiment 7- with population size 20 and 05-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9632 (XGB)	0.9593 (XGB)	0.9226 (XGB)	0.9624 (XGB)	0.9613 (XGB)	0.9226 (XGB)	0.9447 (SVM)	0.9924 (XGB)	1.9404 (KNN)
Experiment 8- with population size 20 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9608 (XGB)	0.9658 (XGB)	0.9266 (XGB)	0.9603 (XGB)	0.9633 (XGB)	0.9266 (XGB)	0.9471 (XGB)	0.9938 (XGB)	3.80728 (KNN)

rationale. The findings from these trials demonstrate that the proposed method successfully differentiates between individuals infected with COVID-19 and those who are healthy. The findings from the experiments demonstrate that this approach enhances classification accuracy for COVID-19 diagnosis, both on its own and when used in conjunction with other methods. The proposed methodology can assist healthcare practitioners in identifying COVID-19 viral infections via CT scans, resulting in improved time efficiency and a higher rate of accurate detection.

Experimentation using GWO algorithm.

8 experiments have been conducted through GWO selected features, the summary of the best results is shown below through Table 2.

Owing to limitations in space, we are showcasing the convergence curves for just one experiment (experiment #1) out of 8 experiments conducted under GWO algorithm umbrella, as illustrated in Fig. 6.

3.5 Experimentation using SSA algorithm

8 experiments have been conducted through SSA selected features, the summary of the best results are shown below through Table 3.

Due to space constraints, we are presenting the convergence curves for only one experiment (experiment #6) out of the eight experiments conducted under the SSA algorithm, as shown in Fig. 7.

Table 3 Best results computed through features returned by SSA algorithms

Experiment -1 with population size 5 and 5-fold cross-validation									
Best value (Classifier Name)	Specific-ity (Spcfty)	Sensitivity (Snstvvty)	Kappa Score(Kappas)	Precision (Prcson)	F1-Score	MCC	Accuracy (Accrcy)	AUC Score (AUCS)	Time in execution (Exctn-time)
	0.9592 (XGB)	0.9585 (XGB)	0.8983 (RF)	0.9585 (XGB)	0.9588 (XGB)	0.9177 (XGB)	0.9443 (KNN)	0.9988 (RF)	4.3644 (KNN)
Experiment -2 with population size 5 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9787 (XGB)	0.9658 (XGB)	0.9363 (XGB)	0.9697 (XGB)	0.9848 (SE)	0.9363 (XGB)	0.9598 (KNN)	0.9958 (XGB)	18.1781 (KNN)
Experiment -3 with population size 10 and 5-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9664 (XGB)	0.9689 (XGB)	0.9274 (XGB)	0.9656 (XGB)	0.9637 (XGB)	0.9274 (XGB)	0.9528 (XGB)	0.9953 (XGB)	2.67372 (KNN)
Experiment 4- with population size 10 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9787 (XGB)	0.9658 (XGB)	0.9363 (XGB)	0.9697 (XGB)	0.9848 (SE)	0.9363 (XGB)	0.9598 (KNN)	0.9958 (XGB)	18.1781 (KNN)
Experiment 5- with population size 15 and 5-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9624 (XGB)	0.9625 (XGB)	0.9258 (XGB)	0.9617 (XGB)	0.9625 (XGB)	0.9258 (XGB)	0.9588 (XGB)	0.9952 (XGB)	6.27671 (KNN)
Experiment 6- with population size 15 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9656 (XGB)	0.9658 (XGB)	0.9386 (XGB)	0.9658 (XGB)	0.9653 (XGB)	0.9386 (XGB)	0.9588 (KNN)	0.9957 (XGB)	11.546 (KNN)
Experiment 7- with population size 20 and 05-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9632 (XGB)	0.9593 (XGB)	0.9226 (XGB)	0.9624 (XGB)	0.9484 (RF)	0.9226 (XGB)	0.9479 (XGB)	0.9951 (XGB)	8.67265 (KNN)
Experiment 8- with population size 20 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9664 (XGB)	0.9658 (XGB)	0.9314 (XGB)	0.9657 (XGB)	0.9658 (XGB)	0.9516 (RF)	0.9584 (RF)	0.9957 (XGB)	17.3342 (KNN)

3.6 Experimentation using hybrid algorithm

8 experiments have been conducted through hybrid selected features, the summary of the best results is shown below through Table 4.

Due to space constraints, we present a comprehensive overview of experiment #2, which involved a population size of 5. This experiment utilized a selected subset of features determined by a hybrid algorithm and employed multiple ML classifiers with 10-fold cross-validation (Table 5).

Owing to limitations in space, we are showcasing the convergence curves for just one experiment (experiment #2) out of 8 experiments conducted under hybrid algorithm umbrella, as illustrated in Fig. 8.

To account for the unequal clinical cost of misclassification, fold-wise confusion matrices were analyzed using 5-fold cross-validation with a population size of 20. Across the five folds, the proposed hybrid GWO–SSA optimized approach in combination with XGBoost classifier consistently produced high true positive counts (ranging from 249 to 251) with very low false negatives (8–9 per fold). Aggregated results show 1251 true positives and only 44 false negatives out of 2482 CT images, demonstrating strong sensitivity and minimizing the risk of missed COVID-19 cases. False positives remain limited (47 cases overall), which is clinically acceptable as such cases can be resolved through confirmatory RT-PCR testing. The low variation in false positives and false negatives across folds confirms the robustness and generalizability of the proposed model, making it suitable for real-world clinical decision-support rather than standalone diagnosis (Fig 9).

The AUC curves resulting from the experiments involving three algorithms, various parameters, and multiple ML classifiers are complex to present comprehensively. Therefore, we have illustrated specific AUC curves (Fig. 10) using a population size of 20 and the hybrid (proposed) algorithm as the FS method.

Table 4 Best results computed through features returned by hybrid algorithms

Experiment -1 with population size 5 and 5-fold cross-validation									
Best value (Classifier Name)	**Specificity (Spcfty)**	**Sensitivity (Snstvvty)**	**Kappa Score (Kappas)**	**Precision (Prcson)**	**F1-Score**	**MCC**	**Accuracy (Accrcy)**	**AUC Score (AUCS)**	**Time in execution (Exctn-time)**
	0.9672 (XGB)	0.9560 (XGB)	0.9234 (XGB)	0.9662 (XGB)	0.9617 (XGB)	0.9234 (XGB)	0.9512 (XGB)	0.9951 (XGB)	4.626 (SVM)
Experiment -2 with population size 5 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9649 (RF)	0.9601 (XGB)	0.9278 (XGB)	0.9578 (XGB)	0.9606 (XGB)	0.9234 (XGB)	0.9509 (XGB)	0.9910 (XGB)	5.102 (KNN)
Experiment -3 with population size 10 and 5-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9646 (RF)	0.9617 (XGB)	0.9289 (XGB)	0.9586 (XGB)	0.9605 (XGB)	0.9210 (XGB)	0.9528 (KNN)	0.9984 (RF)	4.37 (KNN)
Experiment 4- with population size 10 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9632 (XGB)	0.9666 (XGB)	0.9298 (XGB)	0.9627 (XGB)	0.9649 (XGB)	0.9298 (XGB)	0.9560 (RF)	0.9956 (XGB)	7.55 (SVM)
Experiment 5- with population size 15 and 5-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9624 (XGB)	0.9528 (XGB)	0.9153 (XGB)	0.9614 (XGB)	0.9576 (XGB)	0.9153 (XGBoost)	0.9536 (KNN)	0.9946 (XGB)	3.73 (SVM)
Experiment 6- with population size 15 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9648 (XGB)	0.9617 (XGB)	0.9266 (XGB)	0.9641 (XGB)	0.9633 (XGB)	0.9266 (XGB)	0.9552 (KNN)	0.9952 (XGB)	5.22 (KNN)
Experiment 7- with population size 20 and 05-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9648 (XGB)	0.9617 (XGB)	0.9266 (XGB)	0.9641 (XGB)	0.9633 (XGB)	0.9366 (XGB)	0.9463 (XGB)	0.9947 (XGB)	2.33 (KNN)
Experiment 8- with population size 20 and 10-fold cross-validation									
Best Value (Classifier)	**Spcfty**	**Snstvvty**	**Kappas**	**Prcson**	**F1-Score**	**MCC**	**Accrcy**	**AUCS**	**Exctn-time**
	0.9648 (XGB)	0.9674 (XGB)	0.9322 (XGB)	0.9643 (XGB)	0.9661 (XGB)	0.9322 (XGB)	0.9528 (KNN)	0.9949 (XGB)	4.87 (KNN)

3.7 Discussion on overall approach along with significant insights of computed results

The coronavirus has spread quickly around the world over the past few years, and the search for a crucial cure for this disease still continues. Early identification of this disease is crucial for reducing the possibility of increasing death rates and preventing an epidemic spread. Scientists and researchers have employed image processing techniques and sophisticated technology to identify early disease signals in radiological images. To look at methods for early COVID-19 detection, researchers have lately conducted in-depth investigations applying ML and DL models. Furthermore, the selection of suitable features significantly influences the success of ML techniques. Using CT images, the recommended technology based clinical decision support system (CDSS)—whose core is feature extraction and FS techniques—showcased encouraging classification performance, and demonstrated remarkable efficacy in COVID-19 detection, as evidenced by the tables above. Using a FS approach based on soft computing methods significantly enhances classification performance. The improved-sized results show that using FS methods significantly improves classification performance. Furthermore, this approach reduces time-related expenses, which is crucial for effective patient treatment given the present COVID-19 epidemic. This empirical study aims to develop a highly efficient classification system, utilizing a suitable-sized dataset, through the application of image processing engineering, soft computing, and ML and ML-sized methods, improving the ML architecture.

The global COVID-19 epidemic has prompted research on a technology-driven diagnostic approach using CT scans to classify the virus, possibly relieving some of the burden on healthcare professionals. Recent technological advancements in the field of human disease prediction have demonstrated remarkable success. Including FS in ML for COVID-19 patient prediction lowers the waiting time for test findings, enabling healthcare professionals to provide quick and efficient medical treatment. This empirical

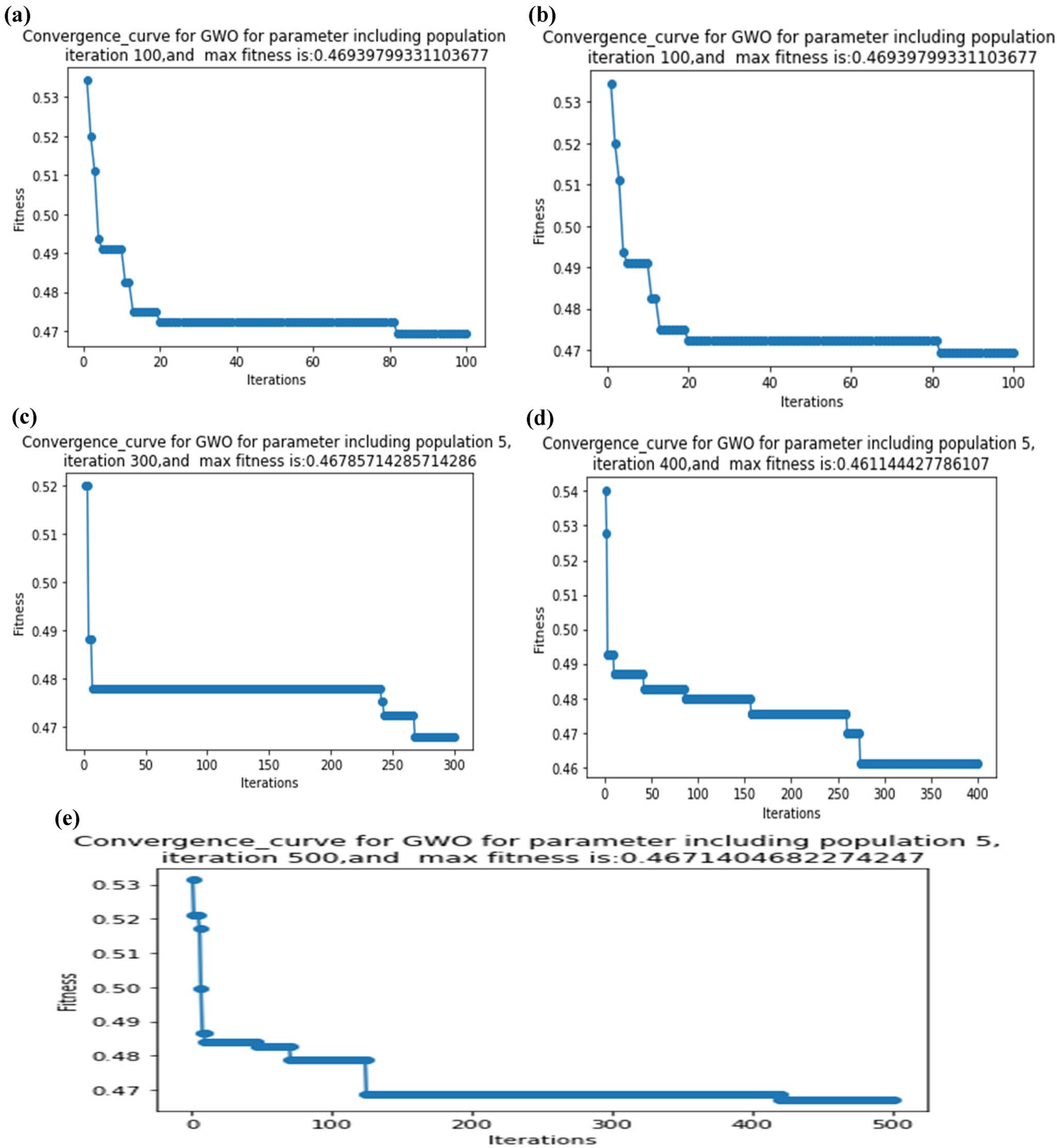

Fig. 6 (**a**) Convergence curve for 100 iterations, (**b**) Convergence curve for 200 iterations, (**c**) Convergence curve for 300 iterations, (**d**) Convergence curve for 400 iterations, (**e**) Convergence curve for 500 iterations (Iteration Vs Fitness) for experiment 1.

research contributes to the development of a highly efficient clinical decision support system that can predict COVID-19 in patients with minimal human intervention and limited processing time. In particular, we focused on the CT scan (an essential tool for the diagnosis of coronavirus disease (COVID-19)) and explored multiple algorithms that could be trained on the COVID-19 data set to develop this model. We efficiently combined two effective nature-inspired feature optimization algorithms, GWO and SSA. Following that, we fed this information (shortlisted features) into several ML models, including SVM, KNN, Random Forest, XGBoost, and a combination of these models (Ensemble

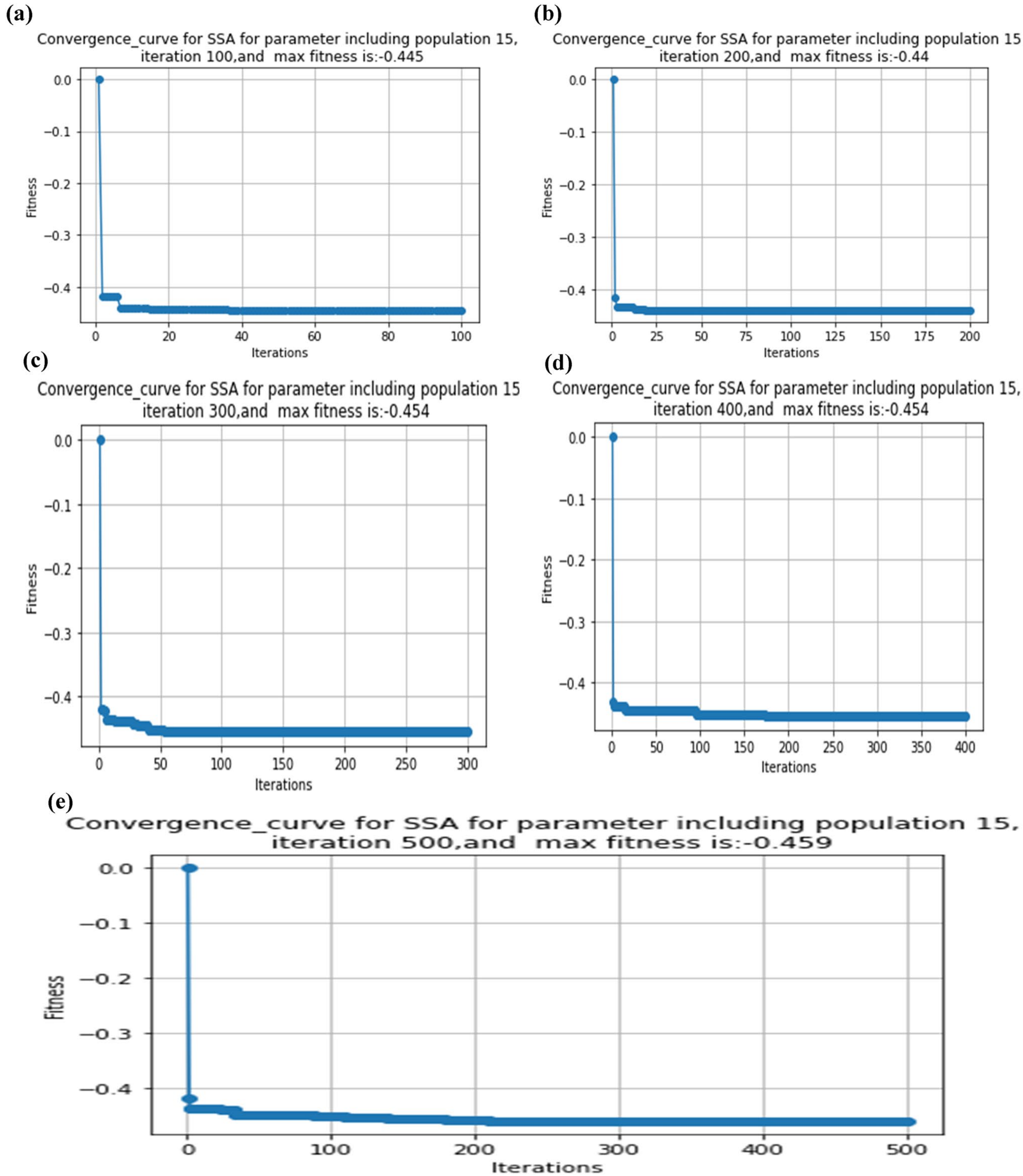

Fig. 7 **(a)** Convergence curve for 100 iterations, **(b)** Convergence curve for 200 iterations, **(c)**Convergence curve for 300 iterations, **(d)**Convergence curve for 400 iterations, **(e)** Convergence curve for 500 iterations (Iteration Vs Fitness) for experiment 6.

model), to precisely estimate the disease. Chest CT images help organize the three main phases of the presented method. Initially, a capable, self-programmed code is developed to implement feature extraction. In the next stage, we identify the most influential features from the original set of extracted features. Different standard ML classifiers are employed to classify data based on relevant features. We assess the performance of the selected algorithms to

Table 5 Table displaying the calculated outcomes of experiment 2

No. of initial features	No of iteration	Population size	Features selected	Fitness value	Features number
213	100	5	[0 2 3 4 6 11 12 13 15 19 22 26 27 29 30 31 34 35 37 38 39 40 43 47 48 56 57 60 61 65 66 67 68 71 72 75 78 80 81 85 86 88 90 91 92 94 96 100 101 107 108 110 111 113 114 115 118 119 122 125 126 127 128 129 130 131 132 135 137 142 145 146 148 150 151 152 153 154 161 163 167 168 169 170 172 174 177 178 182 186 188 189 191 192 193 195 197 199 201 203 206 207 208 211]	0.71041	104
213	200	5	[0 1 2 5 11 12 13 14 15 16 20 21 22 24 25 26 27 37 40 41 42 43 45 51 52 53 55 56 57 58 59 60 61 62 63 65 67 70 72 73 74 77 78 79 86 88 90 91 95 96 99 100 102 103 107 109 110 112 113 114 115 117 118 121 124 126 128 130 131 132 136 138 139 141 142 147 148 149 152 154 156 157 161 162 165 166 169 170 173 174 178 180 181 182 183 185 186 188 190 191 193 195 196 199 201 204 207 208 210 211 212]	0.09650	111
213	300	5	[1 2 3 5 6 7 9 10 11 12 14 15 19 23 24 25 27 28 30 31 34 36 42 44 45 46 48 52 53 55 56 59 60 62 65 66 68 72 73 75 77 78 81 82 84 85 86 90 94 95 98 100 101 102 105 106 107 111 112 114 119 124 128 129 133 136 138 140 141 143 144 145146 148 149 151 153 154 156 158 159 160 161 162 165 167 169 170 171 172 173 177 182 184 186 187 189 190 192 196 199 202 203 205 206 207 209 210 211]	0.01068	109
213	400	5	01 4 5 8 10 11 12 14 15 16 17 18 21 22 23 29 30 31 32 33 36 38 39 40 41 43 45 47 49 51 53 54 56 61 62 63 64 66 69 70 71 72 73 74 75 76 79 85 94 95 97 99 101 106 108 112 113 114 115 117 118 121 123 124 125 126 129 131 135 137 141147 149 151 152 153 156 157 158 160 161 162 164 166 167 168 169 170 171 173 176 178 179 182 184 185 186 187 190 191 193 194 195 197 198 201 203 205 206 207 208 211 212]	0.00132	114
213	500	5	[16 9 11 13 14 15 16 17 21 22 23 25 27 28 29 31 32 33 35 36 42 47 50 51 52 55 58 59 61 63 64 65 67 70 71 75 77 78 80 81 82 83 85 86 90 93 97 98 99 101 102 106 107 108 109 110 111 115 119 120 121 123 124 127 132 133 135 137 139 140 141 143 158 152 153 154 155 156 157 158 159 160 162 165 166 169 170 172 174 175 180 181 184 186 187 198 191 196 199 288 201 202 284 205 206 287 210]	0.00023	108

determine which one yields the highest accuracy and optimal values for the different performance evaluation metrics. Realizing the significance of features reveals how they affect prediction accuracy.

During the trials, we applied both 5- and 10-fold cross-validation methods. The results of these analyses show how well a set of extracted features can detect COVID-19 disease. With the FS approach, we obtained an astonishing accuracy of 96.00%. The proposed system could help medical professionals make decisions, increasing significant time efficiency and a strong success rate in COVID-19 detection with CT images. Reflecting these 24 trials, the computed tables above show that the GWO algorithm reaches a lowest specificity of 0.8905 and a peak specificity of 0.9664 with XGBoost. Using XGBoost, the peak value for the other important factor, sensitivity, came at 0.9658; the minimum came at 0.8559. The minimum recorded was 0.8900; the peak precision attained was 0.9656 (XGBoost). While the minimum recorded score stood at 0.8744, the peak F1 score attained with XGBoost reached 0.9653. We obtained the lowest accuracy of 0.8883 and an amazing 0.9496 (using XGBoost). The AUC ran from 0.9960 using XGBoost to 0.9005. Fascinatingly, XGBoost ranks highest among all the ML models, even though we expected the ensemble method to lead the way. Moreover, a closer inspection of the AUC values shows that we regularly obtained results above 90% in every case, highlighting the extraordinary effectiveness of the suggested approach. The additional metrics, such as the Kappa score and MCC score, stand out among the features identified by the GWO optimization method. The

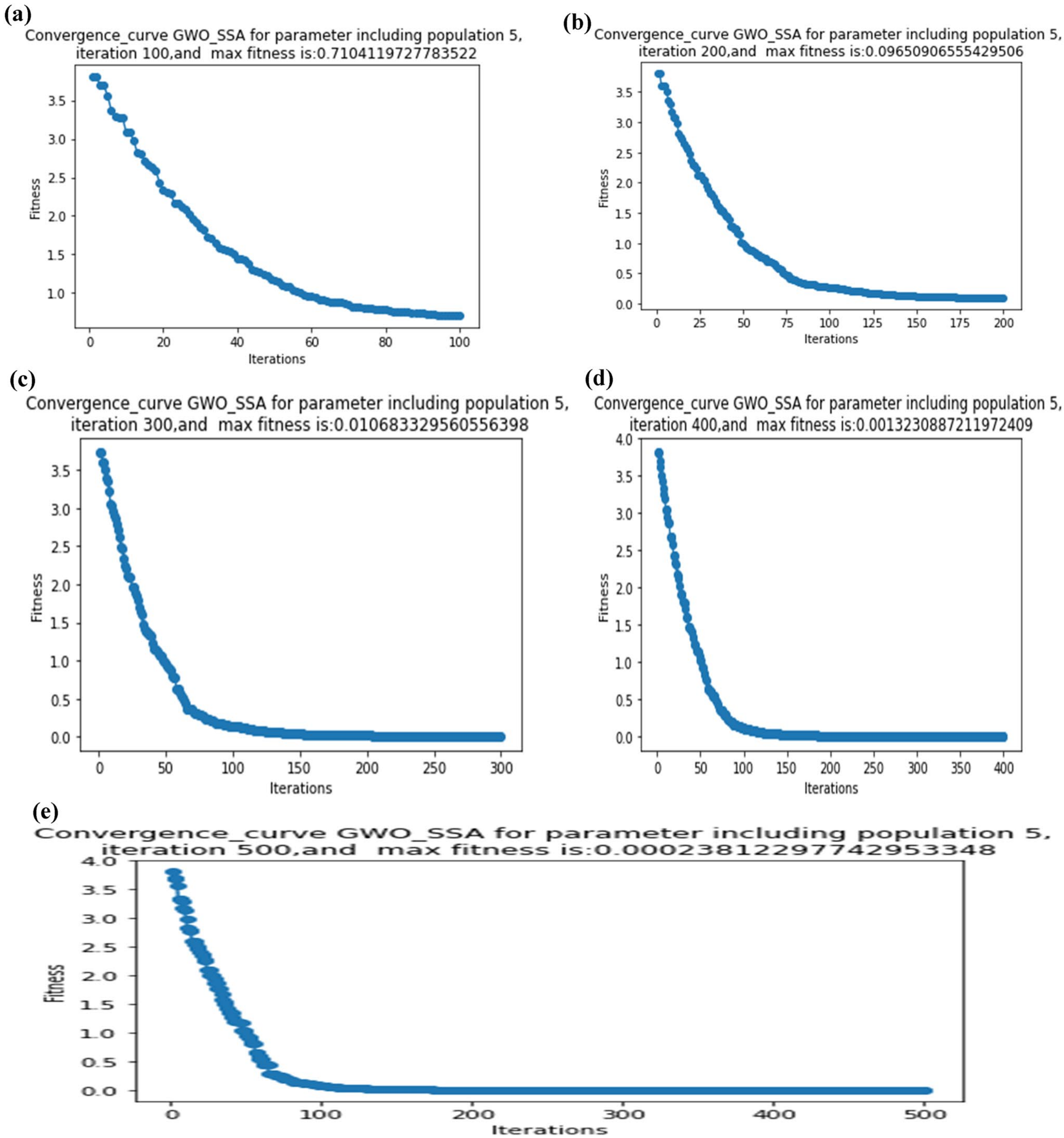

Fig. 8 Convergence curve for 100 iterations, **(b)** Convergence curve for 200 iterations, **(c)** Convergence curve for 300 iterations, **(d)** Convergence curve for 400 iterations, **(e)** Convergence curve(Iteration Vs Fitness) for 500 iterations.

GWO method effectively reduced the original set of 213 features to just 44, achieving an impressive 78% decrease. This achievement demonstrates the algorithm's remarkable success in maintaining the performance of seven efficiency-measuring statistical parameters. The 24 trials shown in the tables above allow one to deduce that for SSA, the minimum recorded specificity is 0.8993, while the maximum specificity attained with XGBoost is 0.9787. The other crucial element falls to a low of 0.8738 and peaks at 0.9689 using XGBoost. Using XGBoost, we observed a peak precision of 0.9697 and a minimum of 0.9022. XGBoost brought the highest F1-score—0.9848—while the lowest F1-score came from 0.8913. With the KNN approach, the accuracy attained an amazing degree of 0.9598; the lowest recorded

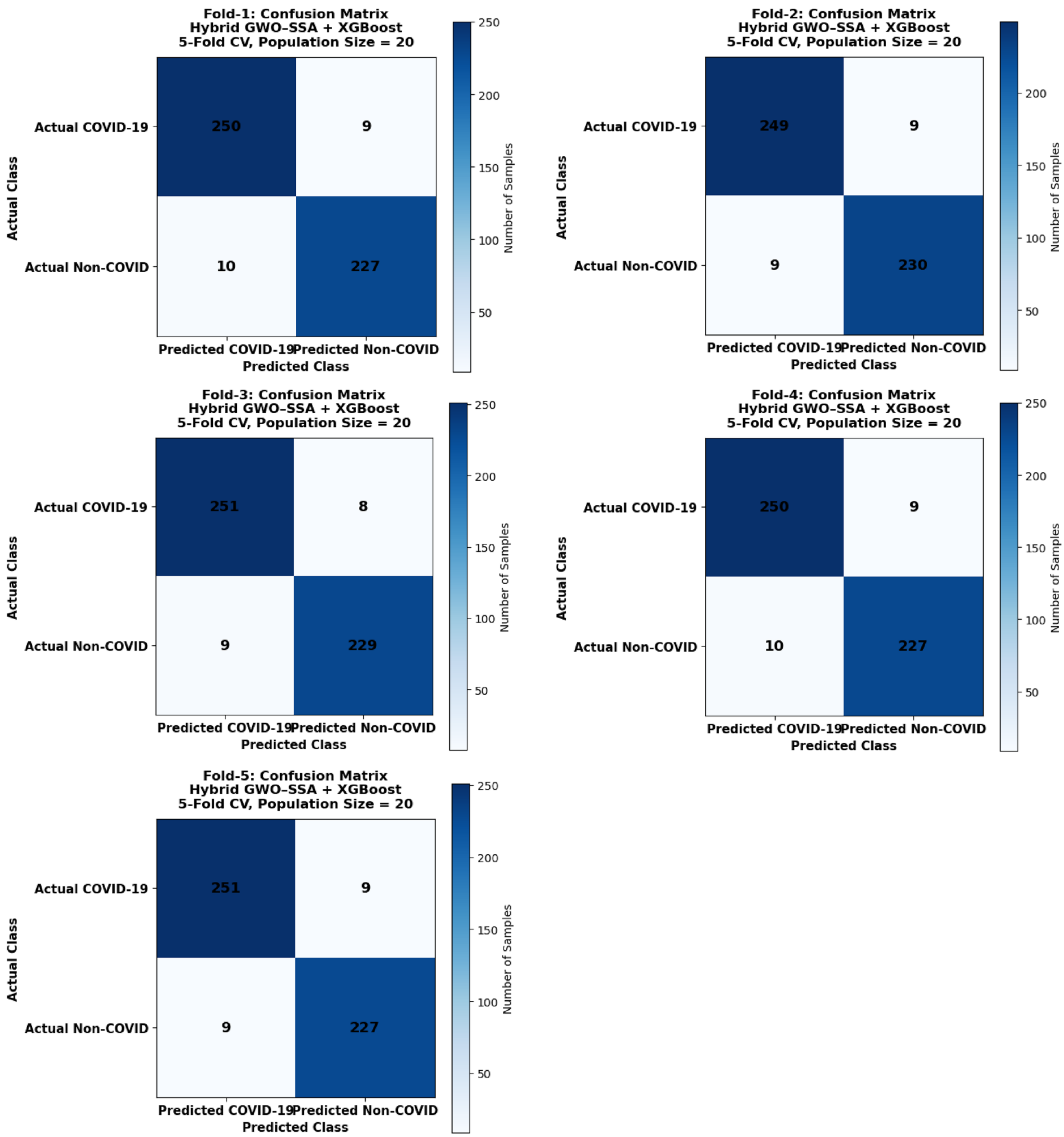

Fig. 9 Fold-wise confusion matrix for hybrid approach case (with population size 20)

accuracy was 0.9008. The AUC ran from 0.9276, using KNN, to 0.9988. Even when taking into account the ensemble prediction, XGBoost and KNN emerge as the most effective ML classifiers in this particular context. Moreover, an analysis of the accuracy and precision measures shows that, in all cases, we regularly obtained results surpassing 90%. This indicates the strong performance of the suggested approach. When weighed against several other measures, including Kappa and MCC scores, the SSA approach displayed remarkable ability. The GWO approach effectively reduced the initial set of 213 features to just 110, demonstrating that we can nearly halve the original set while still maintaining the performance of seven efficiency-measuring statistical parameters.

With XGBoost, the hybrid method achieves a remarkable maximum specificity of 0.9704; the lowest recorded

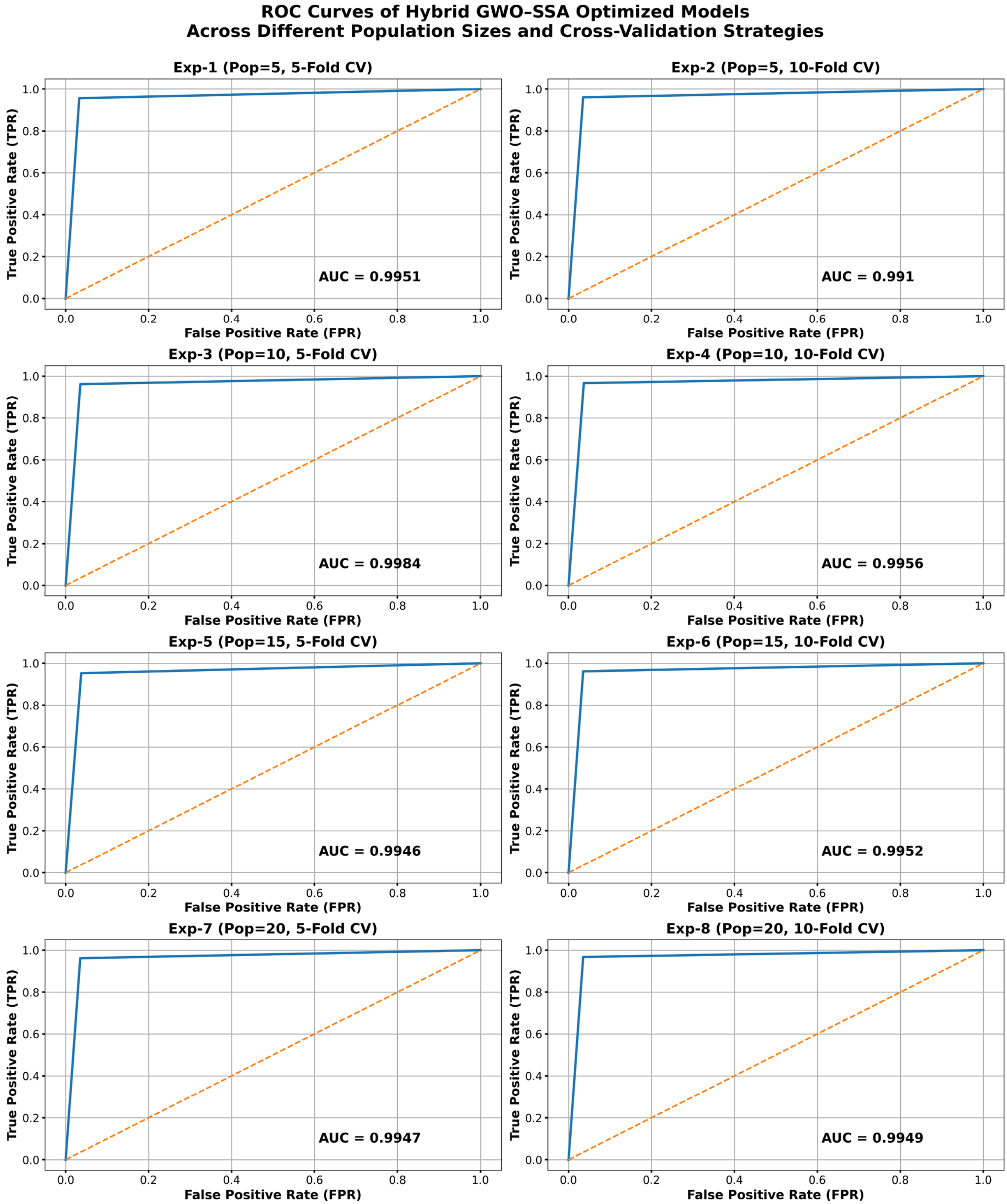

Fig. 10 AUC curves for the Hybrid algorithm case with population size 20.

specificity is 0.9024. Other significant factors' sensitivity peaked at 0.9674 (using XGBoost) and then dropped to 0.8689. XGBoost's best precision was 0.9696; its lowest noted value was 0.9017. Whereas the minimum observed was 0.8915, the peak F1 score attained with XGBoost reached 0.9665. We obtained an outstanding accuracy of 0.9690 by using XGBoost; the lowest accuracy observed was 0.8823. The AUC values, which started at an impressive 0.9999 when using XGBoost, decreased to a minimum of 0.9162. XGBoost breaks the expected outcomes for ensemble techniques, showing itself as the best ML classifier in all three cases. Furthermore, it is evident from the analysis of the precision and AUC values that we regularly obtained at least 90% accuracy, highlighting the remarkable performance of the suggested approach. In addition to the features of the hybrid optimization method, other indicators such as the MCC score and Kappa score demonstrated exceptional performance. This hybrid approach effectively reduced the original set of 213 characteristics to at least 110 features, a reduction of approximately 50%, while maintaining the performance of seven statistical measures used for evaluating efficiency. The previous tables make it clear that the proposed identification model, which used feature extraction and selection methods, was very good at classifying CT images.

3.8 Discussion on influence of computational complexity on practical deployment

This discussion clarifies how computational complexity influences the practical deployment of the GWO, SSA, and Hybrid FS algorithms. Specifically, higher computational complexity increases processing time and resource consumption, which can limit real-time applicability in large-scale or clinical environments. GWO generally offers lower computational overhead due to its simple leader-based updating mechanism, making it more suitable for faster deployment. SSA, may require more iterations to converge, increasing computation time. The GWO–SSA hybrid algorithm has complexity $O(T \times N \times (D + Cf))$ dominated by fitness evaluation (where N: population size, D: dimensionality of the feature space (number of selected variables), T: maximum number of iterations and Cf: cost of evaluating the fitness function (e.g., classifier training + validation)). Optimization ensures practicality, while feature reduction significantly improves real-time inference efficiency, making the approach feasible and clinically deployable. The Hybrid algorithm, though often delivering superior accuracy, can be more computationally intensive because it integrates the operations of both GWO and SSA. These factors have now been explained to help readers understand the trade-off between accuracy and deployment feasibility in practical medical imaging applications.

Practical Note -- Impact on Real Deployment.

In real-world clinical and decision support environments, computational complexity affects:

1. Model Training Time; Higher N or T increases time, but training is offline, so acceptable when accuracy gains are significant.
2. Prediction (Inference) Time: After optimization, the classifier uses fewer features — reducing latency and memory consumption, enabling fast clinical decision-making.
3. Scalability: Because complexity grows linearly with population and feature size, it scales well to large datasets with proper tuning.
4. Deployment Feasibility-While optimization is moderately costly, the final optimized model is lightweight and deployable on hospital servers, cloud systems, and even edge devices.

This study worked on the Grey Wolf Optimization (GWO), Salp Swarm Optimization (SSA), and a hybrid GWO–SSA algorithm with 213 meticulously extracted features. The optimization process is time-consuming and resource-intensive due to the expanding search space with each additional characteristic, population, and iteration. Implementing cross-validation with 5 or 10 folds results in a significantly greater number of training and testing iterations, hence increasing the duration of execution, memory consumption, and CPU load. Since each optimizer must be executed repeatedly for every fold, the computational requirement becomes high, especially for large datasets or systems that demand rapid model development.

Utilizing many classifiers, such as Random Forest, KNN, SVM, XGBoost, or a combination thereof, complicates the mathematical computations significantly. The costs associated with training and generating predictions for each classifier differ. Training KNN is cost-effective, although it requires considerable time for making predictions. As the feature sets expand, the expenses associated with SVM increase. Training Random Forest and XGBoost requires considerable time, and the ensemble approach increases both the model's size, computational complexity and the duration for generating predictions. These factors directly influence the efficacy of operations in real-world scenarios, particularly in areas with constrained computational resources or time restrictions. Selecting the appropriate features will enhance the system's scalability, accelerate inference generation, and facilitate real-world application. This is due to its capacity to simplify the calculations. Therefore, reducing computational complexity through effective FS is

Table 6 Evaluation of our outcomes against current leading approaches

Reference	Type and count of COVID-19 pictures	Accuracy	F1-Score	AUC	Sensitivity	Specificity	Precision
Al-Areqi et al. 2022	746 CT images	98.67	0.9867	--	0.9867	--	0.9867
Basu et al. 2022	2482 CT scan images	97.30	--	--	--	--	--
Panwar et al. 2020	2482 CT scan images	95.61	0.9700	--	0.9500	--	0.9900
Yazdani et al. 2020	2482 CT scan images	92.00	0.9160	0.9700	0.903	--	0.9160
Ibrahim et al. 2023	2482 CT scan images	96.39	09641	--	0.9603	0.9621	0.9680
Ibrahim et al. 2023	2482 CT scan images	97.59	0.9621	--	0.9841	0.9782	0.9688
Goel et al. 2020	2482 CT scan images	97.78	0.9777	0.9788	0.9678	0.9877	0.9877
Wang et al. 2020a, b, c	2482 CT scan images	90.83	0.9087	0.9624	0.8589	--	0.9575
Angelov et al. 2020	2482 CT scan images	88.60	0.8920	0.8860	0.8860		0.8970
Ozyurt et al. 2021	349 COVID-19, 397 healthy CT images	95.84	0.9552	--	--	--	--
Garain et al. 2021	746 patients Chest CT images	--	0.9900	--	--	--	--
Song et al. 2020	201 patients Chest CT images	--	--	0.972	0.8000	0.8500	--
Gifani et al. 2021	800 CT images	85.00	--	--	0.854	--	0.857
Hassantabar et al. 2020	200 CTs of positive patients	93.20	--	--	0.9609	--	--
Sen et al. 2021	2482 CT scans	95.32	0.953	0.953	0.953	--	0.953
Jaiswal et al. 2021	2482chest CT scan dataset	96.25	0.9629	--	0.9629	0.9621	0.9629
Saad et al. 2022	2628 COVID CTs and 1620 Non-COVID	96.13	0.9569		0.9704		0.9437
Our Proposed	2482 CT scans	Maximum 96.90	Maximum 0.9848	Maximum 0.9999	Maximum 0.9689	Maximum 0.9787	Maximum 0.9696

essential for improving scalability, inference speed, and the feasibility of deploying the proposed system in practical, real-world applications.

3.9 Comparison with prior studies and usability

We need to compare the presented approach with other scholars' previous state-of-the-art work, for which we have selected a few of the most relevant recent studies to benchmark our findings, showcasing the effectiveness of our performance. Before we present the comparison table, we must address a few issues. Before conducting the actual comparison, we need to address a number of issues, the first of which is the fact that more than 50% of the studies use X-rays as their images for investigation. Subsequently, various studies examining CT images employed different classification systems: some utilized a 2-class classification, others a 3-class classification, while the remainder implemented a 4-class classification. Moreover, previous studies have rarely selected or used it, making it difficult to compare our findings fairly. The additional issue lies within the methodology. There are very few published studies that incorporate a handcrafted feature extraction technique, followed by selection of the most influential features through an FS approach utilizing soft computing and an ML method for categorization. Most of the previous studies have concentrated on employing DL techniques to identify disease infections, either through direct application, modification of existing models, or deep attributes obtained from intermediate layers of those models. Ultimately, we have identified the studies that meet our criteria, utilizing either our dataset or CT images from an alternative dataset, and focusing on 2-class classification, which aligns with our approach. Prior to analyzing the table that we have constructed and presented here, it is critical to consider all of these factors (Table 6).

3.10 Usability of the suggested approach

Our proposed prediction system is capable of identifying indicators of COVID-19 infection based on the intensity comparison. The volume and dimensions of the dataset utilized, the frequency of trials conducted, and the count of images employed all contributed to generating clear and beneficial outcomes for the medical community. This research introduces a technology-assisted screening approach that leverages CT images to deliver outcomes on par with those of a radiologist. Our proposed model(s) demonstrate a promising classification accuracy that surpasses that of other shortlisted competing models; the alternatives exhibit lower classification accuracy, likely due to their more complex architectures and greater number of training parameters compared to ours. Our proposed designs' capabilities outperform those of the other models we considered. With the help of generated results, this new image processing-soft computing-machine learning-incorporated framework may help identify people who have infections by giving a quick, cheap, and accurate way to diagnose illness. This automated system, with the least amount of human intervention, can recognize complex patterns in medical images. Radiology departments of medical colleges(public or private hospitals) can use the inexpensive model as an effective supplemental tool for image analysis.

4 Conclusion, limitations and future scope

Scholars and the research community have been exploring ways to leverage artificial intelligence (ML and DL) to assist medical professionals in accurately diagnosing COVID-19. It is the responsibility of the researchers to identify an effective, efficient, and precise approach to predict the virus infection and assist humanity in overcoming this pandemic. This study is a step in this direction through which we introduce a precise and intelligent system that classifies medical decisions, potentially assisting healthcare professionals in making informed diagnoses. We have presented a novel clinical decision support framework for COVID-19 viral disease prediction, which applies feature engineering techniques like extraction and selection to chest CT images. The core of this developed automatic detection model employs soft computing algorithms and their hybridized form to recognize the maximum influential features for COVID-19 prediction based on a patient's chest CT images. Initially after preprocessing steps, we extract the features from CT images from a standard benchmark dataset. In the second stage, we employ two well-known as well as one proposed nature-inspired computing-based algorithm(s) to select the most valuable and informative features (attributes). Thus, through this study, we present a soft computing methodology along with a hybridized FS-guided technique aimed at predicting COVID-19 through the analysis of chest CT images. We employ several selected ML techniques in the final phase to determine the most effective classification outcomes (in two categories). Experiments conducted on the COVID-19 benchmark dataset demonstrate that our proposed approach outperforms various recently proposed methods for classifying COVID-19 data using chest CT images. Analyzing and comparing the results indicates that the suggested approach for FS outperforms current techniques across various performance metrics, including AUC. The presented approach attains an AUC of 0.9999, precision of 0.9696, and accuracy of 0.9690, surpassing multiple recent contemporary techniques. The CT dataset used in this study was obtained from Kaggle and may consist of images aggregated from multiple publicly available sources. As a result, it may include variability in patient demographics, COVID-19 severity stages, and imaging protocols. Although the dataset reflects real-world clinical diversity, it may not be fully representative of all populations or uniformly balanced across severity levels. This limitation was considered during analysis, and preprocessing and validation strategies were applied to improve model robustness and generalizability. Although the proposed hybrid GWO–SSA optimized framework demonstrates strong performance on the COVID-19 CT image dataset, several limitations should be acknowledged. First, the optimization process requires iterative population-based searches, which may increase computational time when applied to extremely large datasets or very high-dimensional image features, potentially limiting scalability unless additional parallel or distributed strategies are employed. Second, the model has been validated primarily on a single medical imaging domain, meaning its generalizability to other imaging modalities (such as MRI, X-ray, and ultrasound) or to datasets acquired from different institutions and scanners may be constrained by variations in image quality, class imbalance, and annotation standards. Finally, although the framework is designed to be disease-agnostic, real-world transfer to other chronic disease datasets may still require re-tuning of hyperparameters and re-optimization of features. These limitations highlight the need for future work on cross-dataset validation, computational optimization, and domain adaptation to further enhance robustness and applicability in diverse clinical environments. We dedicated considerable effort to examining the misclassified cases and discovered that insufficient prior COVID data and the substandard quality of certain images might be the primary factors contributing to their misclassification. In our upcoming endeavors, we aim to enhance the system's capacity for precise predictions by integrating various filters and wrapper combinations. Furthermore, we have organized to work together on supplementary COVID datasets. We anticipate creating a mobile web-based tool in the coming years to assist health professionals in diagnosing COVID-19 infections and other conditions. In the future, we will examine the recently introduced soft computing algorithms, aim to integrate them, and evaluate their effects on the outcomes.

Data availability Data will be made available on reasonable request. Data is also freely available and easily downloadable from the internet repository This work is not funded by any external, or internal, or Government or any third party agency.

Declarations

Conflict of interest We, all authors, declare that we do not have any conflict of interest.

References

Aggarwal P, Mishra NK, Fatimah B, Singh P, Gupta A, Joshi SD (2022b) COVID-19 image classification using deep learning:

Advances, challenges and opportunities. Comput Biol Med 144:105350

Aggarwal S, Gupta S, Alhudhaif A, Koundal D, Gupta R, Polat K (2022a) Automated COVID-19 detection in chest X-ray images using fine-tuned deep learning architectures. Expert Syst, 39(3), e12749

Ahemad MT, Hameed MA, Vankdothu R (2022) COVID-19 detection and classification for machine learning methods using human genomic data. Measurement: Sens 24:100537

Al-Areqi F, Konyar MZ (2022) Effectiveness evaluation of different feature extraction methods for classification of covid-19 from computed tomography images: A high accuracy classification study. Biomed Signal Process Control 76:103662

Alkahya MA, Alreahan HO, Algamal ZY (2023) Classification of breast cancer histopathological images using adaptive penalized logistic regression with Wilcoxon Rank Sum Test. Electron J Appl Stat Anal 16(3):507–518

Al-Qaness MA, Ewees AA, Fan H, Aziz AE, M (2020) Optimization method for forecasting confirmed cases of COVID-19 in China. J Clin Med 9(3):674

Al-Thanoon NA, Qasim OS, Algamal ZY (2022) Improving nature-inspired algorithms for feature selection. J Ambient Intell Humaniz Comput 13(6):3025–3035

Angelov P, Soares E (2020) Explainable-by-design approach for covid-19 classification via ct-scan. https://www.medrxiv.org/content/10.1101/2020.04.24.20078584v1

Aravinda CV, Arjun BC, Khan MS, John S, Hussain SA (2025) Explainable AI hybrid CNN-transformer models for enhanced COVID-19 detection in chest X-rays. Netw Model Anal Health Inf Bioinf 14(1):100

Asif S, Awais M, Amjad K, Bilal O, Al-Sabri R, Abdullah M (2024) A deep ensemble learning framework for COVID-19 detection in chest X-ray images. Netw Model Anal Health Inf Bioinf 13(1):30

Barstugan M, Ozkaya U, Ozturk S (2020) Coronavirus (covid-19) classification using ct images by machine learning methods. arXiv preprint arXiv:2003.09424. https://arxiv.org/abs/2003.09424

Basavegowda HS, Dagnew G (2020) Deep learning approach for microarray cancer data classification. CAAI Trans Intell Technol 5(1):22–33

Basu A, Sheikh KH, Cuevas E, Sarkar R (2022) COVID-19 detection from CT scans using a two-stage framework. Expert Syst Appl 193:116377

Brinati D, Campagner A, Ferrari D, Locatelli M, Banfi G, Cabitza F (2020) Detection of COVID-19 infection from routine blood exams with machine learning: a feasibility study. J Med Syst 44(8):1–12

Chen H, Guo J, Wang C, Luo F, Yu X, Zhang W, Zhang Y (2020) Clinical characteristics and intrauterine vertical transmission potential of COVID-19 infection in nine pregnant women: a retrospective review of medical records. lancet 395(10226):809–815

Ewees AA, Al-Qaness MA, Abualigah L, Algamal ZY, Oliva D, Yousri D, Elaziz MA (2023) Enhanced feature selection technique using slime mould algorithm: A case study on chemical data. Neural Comput Appl 35(4):3307–3324

Fang Y, Zhang H, Xie J, Lin M, Ying L, Pang P, Ji W (2020) Sensitivity of chest CT for COVID-19: comparison to RT-PCR. Radiology 296(2):E115–E117. https://pubs.rsna.org/doi/full/10.1148/radiol.2020200432

Farid AA, Selim GI, Khater HAA (2020) A novel approach of CT images feature analysis and prediction to screen for corona virus disease (COVID-19)

Garain A, Basu A, Giampaolo F, Velasquez JD, Sarkar R (2021) Detection of COVID-19 from CT scan images: A spiking neural network-based approach. Neural Comput Appl 33(19):12591–12604

Goel C, Kumar A, Dubey SK, Srivastava V (2020) Efficient deep network architecture for covid-19 detection using computed tomography images. *medRxiv*

Goel T, Murugan R, Mirjalili S, Chakrabartty DK (2021) OptCoNet: an optimized convolutional neural network for an automatic diagnosis of COVID-19. Appl Intell 51:1351–1366

Gopatoti A, Vijayalakshmi P (2022) CXGNet: A tri-phase chest X-ray image classification for COVID-19 diagnosis using deep CNN with enhanced grey-wolf optimizer. Biomed Signal Process Control 77:103860

Hassantabar S, Ahmadi M, Sharifi A (2020) Diagnosis and detection of infected tissue of COVID-19 patients based on lung X-ray image using convolutional neural network approaches. Chaos Solitons Fractals 140:110170

Hemdan EED, Shouman MA, Karar ME (2020) Covidx-net: A framework of deep learning classifiers to diagnose covid-19 in x-ray images. arXiv preprint arXiv:2003.11055. https://arxiv.org/abs/2003.11055

He X, Yang X, Zhang S, Zhao J, Zhang Y, Xing E, Xie P (2020) Sample-efficient deep learning for COVID-19 diagnosis based on CT scans. medrxiv, 2020–2004

Huang C, Wang Y, Li X, Ren L, Zhao J, Hu Y, Cao B (2020) Clinical features of patients infected with 2019 novel coronavirus in Wuhan. China lancet 395(10223):497–506

Ibrahim MR, Youssef SM, Fathalla KM (2023) Abnormality detection and intelligent severity assessment of human chest computed tomography scans using deep learning: a case study on SARS-COV-2 assessment. J Ambient Intell Humaniz Comput 14(5):5665–5688. https://link.springer.com/article/10.1007/s12652-021-03282-x

Jaiswal A, Gianchandani N, Singh D, Kumar V, Kaur M (2021) Classification of the COVID-19 infected patients using DenseNet201 based deep transfer learning. J Biomol Struct Dyn 39(15):5682–5689. https://www.tandfonline.com/doi/abs/10.1080/07391102.2020.1788642

Kahya MA, Altamir SA, Algamal ZY (2020) Improving whale optimization algorithm for feature selection with a time-varying transfer function. Numer Algebra Control Optim 11(1):87–98

Kathamuthu ND, Subramaniam S, Le QH, Muthusamy S, Panchal H, Sundararajan SCM, Zahra MMA (2023) A deep transfer learning-based convolution neural network model for COVID-19 detection using computed tomography scan images for medical applications. Adv Eng Softw 175:103317

Khanna M, Agarwal A, Singh LK, Thawkar S, Khanna A, Gupta D (2023) Radiologist-level two novel and robust automated computer-aided prediction models for early detection of COVID-19 infection from chest X-ray images. Arab J Sci Eng 48(8):11051–11083

Kovács A, Palásti P, Veréb D, Bozsik B, Palkó A, Kincses ZT (2021) The sensitivity and specificity of chest CT in the diagnosis of COVID-19. Eur Radiol 31(5):2819–2824

Li L, Qin L, Xu Z, Yin Y, Wang X, Kong B, Xia J (2020a) Using artificial intelligence to detect COVID-19 and community-acquired pneumonia based on pulmonary CT: evaluation of the diagnostic accuracy. Radiology 296(2):E65–E71. https://pubs.rsna.org/doi/full/10.1148/radiol.2020200905

Li Q, Feng W, Quan YH (2020b) Trend and forecasting of the COVID-19 outbreak in China. *Journal of Infection*, *80*(4), 469-496

Loey M, Manogaran G, Khalifa NEM (2025) A deep transfer learning model with classical data augmentation and CGAN to detect COVID-19 from chest CT radiography digital images. Neural Comput Appl 37(35):29099–29111. https://link.springer.com/article/10.1007/s00521-020-05437-x

Maghdid HS, Asaad AT, Ghafoor KZ, Sadiq AS, Mirjalili S, Khan MK (2021) April Diagnosing COVID-19 pneumonia from X-ray and CT images using deep learning and transfer learning algorithms.

In Multimodal image exploitation and learning 2021 (Vol. 11734, pp. 99-110). SPIE
Mirjalili S, Gandomi AH, Mirjalili SZ, Saremi S, Faris H, Mirjalili SM (2017) Salp Swarm Algorithm: A bio-inspired optimizer for engineering design problems. Adv Eng Softw 114:163–191
Mirjalili S, Mirjalili SM, Lewis A (2014) Grey wolf optimizer. Adv Eng Softw 69:46–61
Mohamadou Y, Halidou A, Kapen PT (2020) A review of mathematical modeling, artificial intelligence and datasets used in the study, prediction and management of COVID-19. Appl Intell 50(11):3913–3925
Narin A, Kaya C, Pamuk Z (2021) Automatic detection of coronavirus disease (covid-19) using x-ray images and deep convolutional neural networks. Pattern Anal Appl 24(3):1207–1220. https://link.springer.com/article/10.1007/s10044-021-00984-y
Ozyurt F, Tuncer T, Subasi A (2021) An automated COVID-19 detection based on fused dynamic exemplar pyramid feature extraction and hybrid feature selection using deep learning. Comput Biol Med 132:104356
Panwar H, Gupta PK, Siddiqui MK, Morales-Menendez R, Bhardwaj P, Singh V (2020) A deep learning and grad-CAM based color visualization approach for fast detection of COVID-19 cases using chest X-ray and CT-Scan images. Chaos Solitons Fractals 140:110190
PlamenEduardo (2020) SARS-COV-2 ct-scan dataset. [Online]. https://www.kaggle.com/plameneduardo/sarscov2-ctscan-dataset. (Accessed: 31-March-2022)
Qasim OS, Algamal ZY (2021) A gray wolf algorithm for feature and parameter selection of support vector classification. Int J Comput Sci Math 13(1):93–102
Rahman T, Khandakar A, Abir FF, Faisal MAA, Hossain MS, Podder KK, Chowdhury ME (2022) QCovSML: A reliable COVID-19 detection system using CBC biomarkers by a stacking machine learning model. Comput Biol Med 143:105284
Rehman A, Naz S, Khan A, Zaib A, Razzak I (2022) April Improving coronavirus (COVID-19) diagnosis using deep transfer learning. In Proceedings of International Conference on Information Technology and Applications: ICITA 2021 (pp. 23-37). Singapore: Springer Nature Singapore
Saad W, Shalaby WA, Shokair M, El-Samie FA, Dessouky M, Abdellatef E (2022) COVID-19 classification using deep feature concatenation technique. J Ambient Intell Humaniz Comput 13(4):2025–2043
Sahin ME (2022) Deep learning-based approach for detecting COVID-19 in chest X-rays. Biomed Signal Process Control 78:103977
Sahlol AT, Yousri D, Ewees AA, Al-Qaness MA, Damasevicius R, Elaziz MA (2020) COVID-19 image classification using deep features and fractional-order marine predators algorithm. Sci Rep 10(1):1–15
Salama GM, Mohamed A, Abd-Ellah MK (2024) COVID-19 classification based on a deep learning and machine learning fusion technique using chest CT images. Neural Comput Appl 36(10):5347–5365
Sen S, Saha S, Chatterjee S, Mirjalili S, Sarkar R (2021) A bi-stage feature selection approach for COVID-19 prediction using chest CT images. Appl Intell 51(12):8985–9000
Shaban WM, Rabie AH, Saleh AI, Abo-Elsoud MA (2020) A new COVID-19 Patients Detection Strategy (CPDS) based on hybrid feature selection and enhanced KNN classifier. Knowl Based Syst 205:106270
Shahin OR, Alshammari HH, Taloba AI Abd El-Aziz RM Machine learning approach for autonomous detection and classification of COVID-19 virus. Comput Electr Eng, 101, 108055
Gifani P, Shalbaf A, Vafaeezadeh M (2021) Automated detection of COVID-19 using ensemble of transfer learning with deep convolutional neural network based on CT scans. Int J Comput Assist Radiol Surg 16(1), 115–123. https://link.springer.com/article/10.1007/s11548-020-02286-w
Shrivastava K, Khanna M (2025) CFA–ASA: An Adaptive Hybrid Feature Selection Approach for Thyroid Disease Prediction. SN Comput Sci 6(7):781
Singh D, Kumar V, Kaur M (2020) Classification of COVID-19 patients from chest CT images using multi-objective differential evolution–based convolutional neural networks. Eur J Clin Microbiol Infect Dis 39(7):1379–1389
Singh LK, Khanna M (2022) A novel multimodality based dual fusion integrated approach for efficient and early prediction of glaucoma. Biomed Signal Process Control 73:103468
Soares E, Angelov, Plamen B, Froes SH, Michele, Abe K (2020) Daniel. SARS-CoV-2 CT-scan dataset: A large dataset of real patients CT scans for SARS-CoV-2 identification. medRxiv. https://doi.org/10.1101/2020.04.24.20078584
Song J, Wang H, Liu Y, Wu W, Dai G, Wu Z, Deng K (2020) End-to-end automatic differentiation of the coronavirus disease 2019 (COVID-19) from viral pneumonia based on chest CT. Eur J Nucl Med Mol Imaging 47(11):2516–2524
Tang Z, Zhao W, Xie X, Zhong Z, Shi F, Liu J, Shen D (2020) Severity assessment of coronavirus disease 2019 (COVID-19) using quantitative features from chest CT images. arXiv preprint arXiv:2003.11988. https://arxiv.org/abs/2003.11988
Tingting Y, Junqian W, Lintai W, Yong X (2019) Three-stage network for age estimation. CAAI Trans Intell Technol 4(2):122–126
Wang L, Lin ZQ, Wong A (2020c) Covid-net: A tailored deep convolutional neural network design for detection of covid-19 cases from chest x-ray images. Sci Rep 10(1):1–12
Wang S, Kang B, Ma J, Zeng X, Xiao M, Guo J, Xu B (2021) A deep learning algorithm using CT images to screen for Corona Virus Disease (COVID-19). Eur Radiol 31(8):6096–6104
Wang S, Kang B, Ma J, Zeng X, Xiao M, Guo J, Cai M, Yang J, Li Y, Meng X, Xu B (2021) A deep learning algorithm using CT images to screen for Corona Virus Disease (COVID-19). Eur Radiol 31:6096–6104
Wang S, Zha Y, Li W, Wu Q, Li X, Niu M, Tian J (2020a) A fully automatic deep learning system for COVID-19 diagnostic and prognostic analysis. Eur Respir J 56(2). https://doi.org/10.1183/13993003.00775-2020
Wang Z, Liu Q, Dou Q (2020b) Contrastive cross-site learning with redesigned net for COVID-19 CT classification. IEEE J Biomedical Health Inf 24(10):2806–2813
Xu X, Jiang X, Ma C, Du P, Li X, Lv S, Li L (2020) A deep learning system to screen novel coronavirus disease 2019 pneumonia. Engineering 6(10):1122–1129
Yazdani S, Minaee S, Kafieh R, Saeedizadeh N, Sonka M (2020) Covid ct-net: Predicting covid-19 from chest ct images using attentional convolutional network. arXiv preprint arXiv:2009.05096
Zhang J, Xie Y, Pang G, Liao Z, Verjans J, Li W, Xia Y (2020(a)).Viral pneumonia screening on chest X-rays using confidence-aware anomaly detection.IEEE transactions on medical imaging, 40(3), 879–890
Zhang M, Wu Q, Chen H, Heidari AA, Cai Z, Li J, Mansour RF (2023) Whale optimization with random contraction and Rosenbrock method for COVID-19 disease prediction. Biomed Signal Process Control 83:104638